arXiv:2512.24610v2 [math.AP] 27 Sep 2026

HALF-SPACE MINIMIZING SOLUTIONS OF A TWO-DIMENSIONAL ALLEN-CAHN SYSTEM

ZHIYUAN GENG

Abstract. This paper studies minimizing solutions to a two-dimensional Allen-Cahn system on the upper half-plane, subject to Dirichlet boundary conditions,

$$\Delta u - \nabla_u W(u) = 0, \quad u : \mathbb{R}^2_+ \to \mathbb{R}^2, \ u = u_0 \text{ on } \partial\mathbb{R}^2_+,$$

where $W : \mathbb{R}^2 \to [0, \infty)$ is a multi-well potential. We give a complete classification of such half-space minimizing solutions in terms of their blow-down limits at infinity. In addition, we characterize the asymptotic behavior of solutions near the associated sharp interfaces.

1. Introduction

1.1. **Mathematical formulation and main results.** In this paper, we investigate minimizing solutions to the system

$$\Delta u - \nabla_u W(u) = 0, \quad u : \overline{\mathbb{R}^2_+} \to \mathbb{R}^2, \tag{1.1}$$

where $\mathbb{R}^2_+$ is the upper half-plane $\{(x, y) \in \mathbb{R}^2 : y > 0\}$. Here $W : \mathbb{R}^2 \to \mathbb{R}_+ \cup \{0\}$ is a nonnegative potential function that vanishes at finitely many points in $\mathbb{R}^2$. (1.1) is the Euler-Lagrange equation corresponding to the energy functional

$$J(u, \Omega) := \int_\Omega \left(\frac{1}{2}|\nabla u|^2 + W(u) \right) dz, \quad \forall \text{ bounded Lipschitz domain } \Omega \subset \mathbb{R}^2_+. \tag{1.2}$$

We define *minimizing solutions* in the sense of De Giorgi as follows.

Definition 1.1. *A function $u \in H^1_{loc}(\overline{\mathbb{R}^2_+}; \mathbb{R}^2)$ is a* minimizing solution *of* (1.1) *if, for every bounded Lipschitz domain $\Omega \subset \mathbb{R}^2_+$ and every $v \in H^1(\Omega; \mathbb{R}^2)$ satisfying $u|_\Omega - v \in H^1_0(\Omega; \mathbb{R}^2)$, one has*

$$J(u, \Omega) \le J(v, \Omega). \tag{1.3}$$

We also call u a local minimizer of the functional (1.2).

We give the following hypotheses on the potential function W. For the hypotheses (H_3) and (H_3'), we assume that one of them holds.

(H_1) $W \in C^3(\mathbb{R}^2; [0, +\infty))$, $\{z \in \mathbb{R}^2 : W(z) = 0\} = \{a_1, a_2, ..., a_N\}$, $\nabla_u W(u) \cdot u > 0$ if $|u| > M$, and

$$c_1|\xi|^2 \le \xi^T \nabla^2 W(a_i) \xi \le c_2|\xi|^2, \quad i \in \{1, ..., N\},$$

where M, c_1, c_2 are positive constants.

Department of Mathematical Sciences, University of Arkansas, 850 West Dickson Street, Fayetteville, AR 72701

E-mail address: zgeng@uark.edu.

Date: September 29, 2026.

Key words and phrases. half-space solution, Allen-Cahn system, minimizing partition, blow-down limit, diffuse interface.

The author was partially supported by AMS-Simons Travel Grant #25014588.

(H_2) For $i \neq j$, $i, j \in \{1, ..., N\}$, there exists a minimizing heteroclinic connection $U_{ij} \in H^1_{loc}(\mathbb{R}; \mathbb{R}^2)$ that minimizes the following 1D energy:

$$J_1(U) := \int_{\mathbb{R}} \left(\frac{1}{2}|U'|^2 + W(U) \right) d\eta, \quad \lim_{\eta \to -\infty} U(\eta) = a_i, \ \lim_{\eta \to +\infty} U(\eta) = a_j.$$

Denote by σ_{ij} the minimal energy $J_1(U_{ij})$. $\sigma_{ij} = \sigma_{ji}$ for any pair $i \neq j$.

(H_3) $N = 3$ and the following triangle inequality holds:

$$\sigma_{ij} + \sigma_{jk} > \sigma_{ik}, \quad \forall \{i, j, k\} = \{1, 2, 3\}. \tag{1.4}$$

We refer to [4, Proposition 2.6 & Proposition 2.12] for the sufficiency of (1.4) for the existence of all three U_{ij}.

(H'_3) $N > 3$ and $\sigma_{ij} \equiv \sigma$ for any $i, j \in \{1, ..., N\}$, $i \neq j$.

A Dirichlet boundary condition will be imposed on $\partial \mathbb{R}^2_+$:

$$u = u_0 \ \text{ on } \partial \mathbb{R}^2_+ = \{(x, 0) : x \in \mathbb{R}\}, \tag{1.5}$$

where $u_0 \in C^3(\mathbb{R}, \mathbb{R}^2)$. We further assume that u_0 satisfies the following:

$$\lim_{x \to \infty} u_0(x) = a_1, \quad \lim_{x \to -\infty} u_0(x) = a_2.$$

$$\int_{-\infty}^{\infty} \left(\frac{1}{2}|u_0'|^2 + W(u_0) \right) dx < \infty. \tag{1.6}$$

The energy bound (1.6) together with Hypothesis (H_1) implies that

$$\int_0^{\infty} |u_0(x) - a_1|^2 \, dx + \int_{-\infty}^{0} |u_0(x) - a_2|^2 \, dx < \infty. \tag{1.7}$$

We note that it is easy to find boundary data u_0 satisfying all these conditions. For example, one can choose any function converging to a_1, a_2 at an exponential rate $e^{-k|x|}$, with the same type of decay on its derivative.

We also assume that u and $|\nabla u|$ are uniformly bounded. Specifically, there exists a constant $M > 0$ such that

$$\|u\|_{L^\infty(\mathbb{R}^2_+; \mathbb{R}^2)} + \|\nabla u\|_{L^\infty(\mathbb{R}^2_+; \mathbb{R}^{2\times 2})} \leq M. \tag{1.8}$$

In fact, uniform boundedness of $|\nabla u|$ follows from uniform boundedness of $|u|$, C^2 regularity of u_0, and standard elliptic regularity argument up to boundary. We state (1.8) in this form for convenience.

We also use the following degenerate Riemannian metric on $\mathbb{R}^2$ induced by the Allen-Cahn energy,

$$d(p, q) := \inf \left\{ \sqrt{2} \int_0^1 |\gamma'(t)| \sqrt{W(\gamma(t))} \, dt : \gamma \in C^1([0, 1], \mathbb{R}^2), \, \gamma(0) = p, \, \gamma(1) = q \right\}, \quad \forall p, q \in \mathbb{R}^2. \tag{1.9}$$

Intuitively, $d(p, q)$ quantifies the minimal energy needed to move from p to q in the phase plane. It is worth mentioning that $d(a_i, a_j) = \sigma_{ij}$.

The minimizing solution u of (1.1) subject to (1.5) is a diffuse analog of a half-space minimizing partition. For $v \in \mathrm{BV}_{loc}(\overline{\mathbb{R}^2_+}; \{a_1, ..., a_N\})$ and for every bounded Lipschitz domain $\Omega \subset \mathbb{R}^2_+$, define

$$\begin{aligned} J_*(v, \Omega) := & \sum_{1 \le i < j \le N} \sigma_{ij} \mathcal{H}^1(\partial^* v^{-1}(a_i) \cap \partial^* v^{-1}(a_j) \cap \Omega) \\ & + \int_{\partial^* \Omega \cap \partial \mathbb{R}^2_+ \cap \{x>0\}} d(\mathrm{tr}_\Omega\, v, a_1)\, d\mathcal{H}^1 + \int_{\partial^* \Omega \cap \partial \mathbb{R}^2_+ \cap \{x<0\}} d(\mathrm{tr}_\Omega\, v, a_2)\, d\mathcal{H}^1. \end{aligned} \tag{1.10}$$

Here ∂^* denotes the reduced boundary, and $\mathrm{tr}_\Omega\, v$ denotes the interior trace of v on $\partial\Omega$.

Definition 1.2. *$v \in \mathrm{BV}_{loc}(\overline{\mathbb{R}^2_+}; \{a_1, ..., a_N\})$ is called a (Dirichlet)* minimizing partition map *if, for every bounded Lipschitz domain $\Omega \subset \mathbb{R}^2_+$ and every $w \in \mathrm{BV}(\Omega; \{a_1, ..., a_N\})$ satisfying*

$$\mathrm{tr}_\Omega v = \mathrm{tr}_\Omega w, \qquad \mathcal{H}^1\text{-a.e. on } \partial\Omega \cap \mathbb{R}^2_+,$$

one has

$$J_*(v, \Omega) \le J_*(w, \Omega).$$

Define the blow-down map of u at the scale $r \gg 1$ by

$$u_r(z) := u(rz), \quad z \in \mathbb{R}^2_+.$$

Then u_r is a local minimizer of the functional

$$J_r(u, \Omega) := \int_\Omega \left(\frac{1}{2r} |\nabla u|^2 + rW(u) \right) dz,$$

subject to the boundary condition

$$u_r(x, 0) = u_0(rx), \quad \forall x \in \mathbb{R}.$$

Our first result classifies the minimizing solution u in terms of its asymptotic behavior at infinity by studying the limit of u_r, which can be characterized by minimal cones associated with the functional (1.10).

Theorem 1.3. *Assume (H_1), (H_2) and either (H_3) or (H'_3). Let $u : \mathbb{R}^2_+ \to \mathbb{R}^2$ be a minimizing solution of (1.1) satisfying (1.5)–(1.8). There exists a minimizing partition map $\tilde{u} \in \mathrm{BV}_{loc}(\overline{\mathbb{R}^2_+}; \{a_1, a_2, ..., a_N\})$ in the sense of Definition 1.2 such that $\nabla \tilde{u} \cdot z = 0$ in the distributional sense, and*

$$\lim_{r \to \infty} \|u_r - \tilde{u}\|_{L^1(K)} = 0, \ \forall K \Subset \mathbb{R}^2_+,$$

where $u_r(z) := u(rz)$. In polar coordinates $z = (r\cos\theta, r\sin\theta)$, extend the definition of $\tilde{u}$ to $\partial\mathbb{R}^2_+$ by setting

$$\hat{u}(r, \theta) = \begin{cases} \tilde{u}(r, \theta), & 0 < \theta < \pi, \\ a_1, & \theta = 0, \\ a_2, & \theta = \pi. \end{cases} \tag{1.11}$$

Then there exist $0 \le \alpha_1 \le \alpha_2 \le \pi$ with the following properties. If $\alpha_1 < \alpha_2$, there is an index $i \in \{3, ..., N\}$ such that

$$\hat{u}(r, \theta) = \begin{cases} a_i, & \alpha_1 < \theta < \alpha_2, \\ a_1, & 0 \le \theta \le \alpha_1, \\ a_2, & \alpha_2 \le \theta \le \pi. \end{cases}$$

Otherwise if $\alpha_1 = \alpha_2$, *then*

$$\hat{u}(r,\theta) = \begin{cases} a_1, & 0 \le \theta < \alpha_1, \\ a_2, & \alpha_2 < \theta \le \pi. \end{cases}$$

Consequently, $\tilde{u}$ *belongs to one of the following categories:*

a) Constant: $\alpha_1 = \alpha_2 = 0$, $\alpha_1 = \alpha_2 = \pi$ *or* $\alpha_1 = 0$, $\alpha_2 = \pi$.
b) Two-phase profile: $0 < \alpha_1 < \alpha_2 = \pi$, $0 = \alpha_1 < \alpha_2 < \pi$, *or* $0 < \alpha_1 = \alpha_2 < \pi$.
c) Triple junction: $0 < \alpha_1 < \alpha_2 < \pi$.

Moreover, whenever $\alpha_1 < \alpha_2$, *it holds that* $\alpha_2 - \alpha_1 \ge \frac{2\pi}{3}$ *when* (H_3') *holds and* $\alpha_2 - \alpha_1 \ge \theta_3$ *when* (H_3) *holds, in which case* $i = 3$. *Here* θ_3 *is the angle determined by Young's law:*

$$\frac{\sin\theta_1}{\sigma_{23}} = \frac{\sin\theta_2}{\sigma_{13}} = \frac{\sin\theta_3}{\sigma_{12}}, \qquad \sum_{i=1}^{3} \theta_i = 2\pi,\ \theta_i \in (0,\pi). \tag{1.12}$$

Remark 1.1. In cases a) and b), since the blow-down limit may not coincide with a_1 on $\{(x,0) : x > 0\}$ or with a_2 on $\{(x,0) : x < 0\}$, a boundary transition layer is allowed. u may transition from a phase a_i to a_1/a_2 within a thin layer near the boundary $\partial\mathbb{R}^2_+$.

For any $0 \le \alpha \le \pi$, let l_α denote the ray emanating from the origin in direction α:

$$l_\alpha := \{(r\cos\alpha, r\sin\alpha) : r > 0\}.$$

Take a constant $\gamma > 0$. We define the diffuse interface as

$$\mathcal{D}_\gamma := \{z \in \mathbb{R}^2_+ : \min_i |u(z) - a_i| \ge \gamma\}. \tag{1.13}$$

Generally speaking, $\mathcal{D}_\gamma$ contains points that do not belong to any phase a_i, which therefore represents the interfacial region that separates coexisting phases. From Theorem 1.3, we know that u resembles $\hat{u}$ at infinity; therefore, it is natural to expect $\mathcal{D}_\gamma$ to be close to the jump set of $\hat{u}$ written as $\mathcal{S}(\hat{u}) = l_{\alpha_1} \cup l_{\alpha_2}$. Below we state our second main result, which confirms this property and refines the profile of u near $\mathcal{S}(\hat{u})$.

Theorem 1.4. *Assume* (H_1), (H_2), *either* (H_3) *or* (H_3'), *and the following extra hypothesis on* W:

(H_4) *For any* $i \ne j$, *the heteroclinic connection* U_{ij} *in* (H_2) *is unique up to translation. Moreover, for the operator* L_{ij} *defined by*

$$L_{ij}v = -v'' + \nabla^2 W(U_{ij}(x))v, \quad v \in H^2(\mathbb{R};\mathbb{R}^2),$$

the kernel is one-dimensional, i.e. $\ker L_{ij} = \operatorname{span}\{\mathrm{U}'_{\mathrm{ij}}\}$.

Let $u : \mathbb{R}^2_+ \to \mathbb{R}^2$ *be a minimizing solution of* (1.1) *satisfying* (1.5)–(1.8). *Take* $\hat{u}$ *as the extended blow-down map of* u *defined in Theorem 1.3 and suppose*

$$\mathcal{S}(\hat{u}) = l_{\alpha_1} \cup l_{\alpha_2}, \qquad \text{for some } \ 0 < \alpha_1 \le \alpha_2 < \pi.$$

For every sufficiently small $\gamma > 0$, *there exists a constant* $C = C(u,\gamma,W)$ *such that for sufficiently large* R,

$$(\mathcal{D}_\gamma \cap \partial B_R) \subset \{(R\cos\theta, R\sin\theta) : \theta \in (\alpha_1 - CR^{-1}, \alpha_1 + CR^{-1}) \cup (\alpha_2 - CR^{-1}, \alpha_2 + CR^{-1})\}. \tag{1.14}$$

Furthermore, suppose for some $\alpha \in (0,\pi)$, $l_\alpha \subset \mathcal{S}(\hat{u})$ *separates the phases* a_i *and* a_j, *with* a_i *on the side* $\theta < \alpha$ *and* a_j *on the side* $\theta > \alpha$. *Set*

$$\mathbf{e}_\alpha := (\cos\alpha, \sin\alpha), \quad \mathbf{e}_\alpha^\perp := (-\sin\alpha, \cos\alpha).$$

Then there exists a constant h *such that for every* $L > 0$ *and every* $\beta \in (0,1)$,

$$\lim_{x\to+\infty} \|u(x\mathbf{e}_\alpha + y\mathbf{e}_\alpha^\perp) - U_{ij}(y-h)\|_{C^{2,\beta}([-L,L];\mathbb{R}^2)} = 0. \tag{1.15}$$

Remark 1.2. The nondegeneracy condition (H_4) on the heteroclinic connection U_{ij} was first explicitly imposed in [38] to control the asymptotics of two-phase diffuse solutions along the sharp interface. It is not needed for Theorem 1.3.

Remark 1.3. (1.14) and (1.15) imply that if there is no boundary sharp interface, $\mathcal{D}_\gamma \cap B_R$ is contained in an $O(1)$ neighborhood of $\bigcup_\alpha l_\alpha$. However, Theorem 1.4 does not cover boundary interfaces, i.e. when $\alpha_i = 0$ or π. The analogous convergence up to the wall to a finite translate of heteroclinic connection is not available. Indeed, suppose $\mathbf{e}_\alpha = (1, 0)$ and $\lim_{x\to+\infty} \|u(x, y) - U_{1j}(y - h)\|_{C^{2,\beta}_{loc}} = 0$, then $\lim_{x\to+\infty} |u_0(x) - a_1| > 0$, which contradicts (1.7).

1.2. **Related results and outline of the paper.** For the scalar Allen-Cahn equation with a double-well potential, the classification of entire solutions is closely related to De Giorgi's famous conjecture [11]: a bounded global solution that is monotone in one direction is expected to be one-dimensional in dimensions $N \le 8$. Numerous important contributions have been made to prove De Giorgi's conjecture and its variants, as well as to understand the relationship between the solutions of the Allen–Cahn equation and minimal surfaces; see for instance, [27, 6, 36, 17, 12, 41, 34, 32, 42, 29, 43, 44, 19] and the expository papers [37, 10] for a detailed account. For the half-space scalar Allen-Cahn equation, solutions are naturally related to the half-space Bernstein theorem for graphical minimal hypersurfaces considered in [15, 13]. Analogous rigidity results for Allen-Cahn were obtained in [16, 31, 14].

In the vector-valued case, minimizing solutions are diffuse analogues of minimizing partitions. The scalar Modica's estimate, and hence the corresponding energy monotonicity formula, are generally unavailable for systems [4, 8]. The convergence of diffuse minimizers to partitions is instead described by Γ-convergence [7, 39, 20], with Dirichlet boundary conditions treated in [24]. For the two-dimensional case, recent developments have shed light into geometric and analytic description of fine structures of the Allen-Cahn solutions. Bethuel [8] studied the critical points of (1.1) and successfully generalized results on the regularity of interfaces from the scalar case to the two-dimensional vectorial case with the help of an asymptotic monotonicity formula. Fusco [23] studied the connectivity of the diffuse interface and showed the existence of a connected optimal network that separates all phases. Alikakos and Fusco [3] investigated two examples with carefully designed Dirichlet boundary data and derived sharp energy lower and upper bounds and pointwise estimates.

Many earlier constructions of non-one-dimensional solutions relied on symmetry. Bronsard, Gui and Schatzman [9] constructed a planar triple-junction solution in an equivariant class for a symmetric three-well potential. Gui and Schatzman [30] constructed a symmetric three-dimensional quadruple-junction solution, while Fusco [22] obtained planar junctions under weaker rotational symmetry assumptions. Other constructions and related variational approaches can be found in [40, 18, 38, 21, 33]. More recently, Alikakos and the author [5] and Sandier and Sternberg [35] independently constructed entire minimizers with triple-junction structure at infinity without symmetry assumptions. In both works, a sequential blow-down limit is a minimizing three partition of the form $\sum_{i=1}^3 a_i \chi_{D_i}$. The uniqueness of this blow-down and the asymptotic heteroclinic profiles along the sharp interfaces were subsequently established in [26, 25].

The half-space solution considered in the present paper (see Definition 1.1) appears naturally as a blow-up map at the boundary phase-transition point of the minimizing solution to $\varepsilon\Delta u - \frac{1}{\varepsilon}\nabla_u W(u) = 0$ on a finite domain Ω, where $u|_{\partial\Omega}$ has a phase segregation structure. The flat boundary changes the sharp-interface problem in the sense that its energy includes the cost of mismatch between the BV trace of a limiting partition and the prescribed phases on the flat boundary. The limiting interface can therefore lie on the boundary as well as in the interior.

Theorem 1.3 shows that every minimizer under our hypotheses has a unique homogeneous blow-down limit, with a constant, two-phase, or three-phase profile. In the three-phase case, the opening angle of the intermediate phase is bounded below by θ_3, which is the angle determined by Young's law (this angle is $2\pi/3$ for equal surface tensions). It needs not equal that angle as in the case of interior junctions. For sharp interfaces strictly inside the half-plane, Theorem 1.4 further establishes uniform distance localization of the diffuse interface and convergence to a single translate of the corresponding heteroclinic connection, under the nondegeneracy hypothesis (H_4).

For the classification of blow-down limits, we develop a substantially simpler and more direct proof than the approaches based on slicing and diffuse-interface localization in [5, 26] and comparison with wetting partitions in [35]. The argument works directly at the level of the limiting partition, combining compactness with geometric comparison to establish homogeneity and classify the possible blow-down profiles.

The paper is organized as follows. In Section 2, a Γ-convergence argument gives both compactness and convergence of the energy measures, including the boundary mismatch term. Section 3 then uses comparison for the limiting partition to obtain a limiting energy density and homogeneity. Comparing this limit with the one-dimensional lower bound on circular slices yields asymptotic energy equipartition and negligible radial energy. Classification of homogeneous minimizing partitions follows from a direct geometric variation and paired calibration arguments.

In Section 4, the divergence-free stress-energy tensor provides a simpler proof for the uniqueness of the blow-down limit, comparing [26]. Its flux across a large semicircle converges to a vector given by a boundary integral and thus independent of the blow-down sequence. And the energy measure convergence and smallness of radial energy help identify the same vector as $\sum_l \sigma_l e_{\alpha_l}$, where α_l are jump angles for a blow-down limit. Thus the jump angles of every sequential blow-down are fixed, which immediately implies uniqueness. Finally, Section 5 obtains a sharp $O(1)$ bound for the energy excess along interior interfaces, which combined with the compactness and coercivity arguments near heteroclinic connections in [38] yields the conclusion of Theorem 1.4.

2. Preliminaries

2.1. **Notation.** We introduce the following notation for use throughout the paper.

- $z = (x, y)$ denotes a point in $\mathbb{R}^2$. The origin $(0, 0)$ is denoted by O.
- $B_r(z)$ is the disk centered at z with radius r. $B_r^+(z) := B_r(z) \cap \mathbb{R}^2_+$. When z is the origin, we simply write B_r and B_r^+.
- Denote by $A^+_{r_1,r_2}(z_0)$ the part of annulus centered at z_0 in $\mathbb{R}^2_+$: $A^+_{r_1,r_2}(z_0) := \{z \in \mathbb{R}^2_+ : r_1 < |z - z_0| < r_2\}$. When z_0 is the origin, we simply write $A^+_{r_1,r_2}$.
- $|\cdot|$ denotes the ordinary modulus of a vector, for example $|u(z) - a_1|$. Denote by $d(\cdot, \cdot)$ the degenerate Riemannian metric on $\mathbb{R}^2$ defined in (1.9).
- $\tilde{u}$ denotes a minimizing partition map in the sense of Definition 1.2.
- ∂_r denotes the radial derivative, ∂_θ denotes the angular derivative, and $\partial_T := \frac{\partial_\theta}{r}$ denotes the counterclockwise unit tangential derivative.
- For $\alpha \in [0, \pi]$, let l_α denote the ray emanating from the origin in the direction α, that is, $l_\alpha := \{(r\cos\alpha, r\sin\alpha) : r > 0\}$.
- Unless otherwise stated, C denotes a constant that depends on the potential W, M in the uniform bound (1.8) and sometimes also on the boundary data u_0. The value of C may vary from line to line.

- For a Lipschitz domain Ω and a partition map $v \in \mathrm{BV}(\Omega; \{a_1, ..., a_N\})$, $\mathrm{tr}_\Omega\, v$ denotes the interior trace of v on $\partial^*\Omega$, see, for instance [28, Chapter 2]. When $\Omega = \mathbb{R}^2_+$, we simply write $\mathrm{tr}\, v$ for the trace on the flat boundary $\partial\mathbb{R}^2_+$.

2.2. **Basic estimates for vector-valued Allen-Cahn minimizers.** First we recall the following 1D energy estimate.

Lemma 2.1 (Lemma 2.3 in [3])**.** *There exist constants $\delta_W > 0$ and $C = C(W) > 0$ such that the following holds. Let $i \neq j \in \{1, ..., N\}$, $\delta < \delta_W$ and $s_- < s_+$ be two real numbers. Let $v \in H^1((s_-, s_+); \mathbb{R}^2)$ satisfy*

$$\max\{|v(s_-) - a_i|, |v(s_+) - a_j|\} \leq \delta.$$

Then

$$J_{(s_-,s_+)}(v) = \int_{s_-}^{s_+} \left(\frac{1}{2}|v'(s)|^2 + W(v(s))\right) ds \geq d(v(s_-), v(s_+)) \geq \sigma_{ij} - C\delta^2.$$

Besides this one-dimensional energy estimate, we also need the following standard pointwise metric estimate, see for instance [7, Proposition 2.1]. Let $\varphi_i(q) := d(q, a_i)$. For every bounded map $u \in H^1 \cap L^\infty(\Omega, \mathbb{R}^2)$, one has

$$|\nabla(\varphi_i \circ u)| \leq \sqrt{2W(u)}|\nabla u| \leq \frac{\varepsilon}{2}|\nabla u|^2 + \frac{1}{\varepsilon}W(u) \qquad \text{a.e. in } \Omega. \tag{2.16}$$

Next we present the vectorial maximum principle for u, which will be useful in our analysis.

Lemma 2.2 (Variational maximum principle, [2])**.** *Assume (H_1), (H_2), and either (H_3) or (H'_3). There exists $r_0 = r_0(W) > 0$ such that the following holds. Let $D \subset \mathbb{R}^2_+$ be a bounded, connected, Lipschitz domain, and let $u \in H^1(D; \mathbb{R}^2) \cap L^\infty(D; \mathbb{R}^2)$ minimize $J(\cdot, D)$ with respect to Dirichlet boundary data. If for some $r < r_0$ and $1 \leq i \leq N$,*

$$|u(z) - a_i| \leq r \quad \mathcal{H}^1\text{-a.e. on } \partial D,$$

then

$$|u(z) - a_i| \leq r, \quad \text{a.e. in } D.$$

We also recall the following vectorial version of the Caffarelli–Córdoba density estimate in [1].

Lemma 2.3 ([1])**.** *Assume (H_1), (H_2), and either (H_3) or (H'_3). Let $u \in W^{1,2}_{loc}(\mathbb{R}^2_+, \mathbb{R}^2)$ be a minimizing solution of* (1.1) *satisfying* (1.8)*. Suppose for some $r_0 > 0$, $\mu > 0$, $0 < \lambda < \min_{i \neq j} |a_i - a_j|$, $1 \leq i \leq N$, $z \in \mathbb{R}^2_+$,*

$$\mathcal{L}^2(B_{r_0}(z) \cap \{|u - a_i| > \lambda\}) \geq \mu, \quad B_{r_0}(z) \subset \mathbb{R}^2_+,$$

then there exists $c = c(W, M, r_0, \lambda, \mu) > 0$ such that

$$\mathcal{L}^2(B_r(z) \cap \{|u - a_i| > \lambda\}) \geq cr^2, \quad \forall\, r \in [r_0, \mathrm{dist}(z, \partial\mathbb{R}^2_+)).$$

The following clearing-out lemma is a direct consequence of Lemma 2.3.

Lemma 2.4. *Assume (H_1), (H_2), and either (H_3) or (H'_3). Let u be a minimizing solution of* (1.1) *satisfying* (1.8)*. There exist constants δ_0, ε_0, r_0, K, k, depending only on W, M, such that if for some i and $r \geq r_0$, $z_0 \in \mathbb{R}^2_+$,*

$$B_r(z_0) \subset \mathbb{R}^2_+, \qquad \fint_{B_r(z_0)} |u - a_i|\, dz \leq \varepsilon_0, \tag{2.17}$$

then

$$\max_{z \in B_{3r/4}(z_0)} |u(z) - a_i| \leq \delta_0 \tag{2.18}$$

Moreover,

$$\max_{z\in B_{r/2}(z_0)} \left(|u(z)-a_i| + |\nabla u| + |\nabla^2 u|\right) \le Ke^{-kr}. \tag{2.19}$$

Proof. Take $\delta_0 \le \frac{1}{2}\min_{i\neq j}|a_i - a_j|$. Suppose for some $z_1 \in B_{3r/4}(z_0)$, $|u(z_1)-a_i| > \delta_0$. Then by (1.8),

$$|u(z)-a_i| > \frac{\delta_0}{2}, \qquad \text{on } B_{\frac{\delta_0}{4M}}(z_1).$$

By Lemma 2.3, there exists $c = c(W,M)$ such that

$$\mathcal{L}^2(B_{r/4}(z_1)\cap\{|u-a_i|>\frac{\delta_0}{2}\}) \ge cr^2,$$

which contradicts (2.17) if we choose ε_0 suitably small. This proves (2.18).

Then from (2.18) and the comparison principle for elliptic equations the exponential decay estimate (2.19) follows (cf. [4, Proposition 6.4]).

□

Next we derive a rough energy upper bound for u on any disk $B_r(z)$.

Lemma 2.5. *Assume* (H_1)*,* (H_2)*, and either* (H_3) *or* (H_3')*. Let u be a minimizing solution of* (1.1) *satisfying* (1.8)*. Then there exists a constant $C = C(W,M)$ such that*

$$J(u, B_r(z)\cap\mathbb{R}^2_+) \le Cr, \quad \forall\, z\in\overline{\mathbb{R}^2_+} \text{ and } r>0. \tag{2.20}$$

Proof. Fix $z_0 \in \overline{R^2_+}$. Let D denote $B_r(z_0)\cap\mathbb{R}^2_+$. By (1.8), $\frac{1}{2}|\nabla u|^2 + W(u) \le C(W,M)$. Therefore when $r\le 2$,

$$J(u,D) \le C\pi r^2 \le Cr.$$

When $r>2$, define

$$\phi(z) := \min\{1, \operatorname{dist}(z,\partial D)\}.$$

Then $\phi(z)$ is 1-Lipschitz and vanishes on ∂D. Set $v\in H^1(D;\mathbb{R}^2)$ as

$$v(z) := (1-\phi(z))u(z) + \phi(z)\,a_1,$$

which is a competitor on D. Both $\nabla v(z)$ and $W(v(z))$ vanish when $\operatorname{dist}(z,\partial D)>1$. We have

$$J(u,D) \le \int_{D\cap\{\operatorname{dist}(z,\partial D)\le 1\}} \left(\frac{1}{2}|\nabla v|^2 + W(v)\right) dz \le C\mathcal{H}^1(\partial D) \le Cr,$$

which completes the proof.

□

Moreover, we present the following compactness result.

Proposition 2.6. *Assume* (H_1)*,* (H_2)*, and either* (H_3) *or* (H_3')*. Let $u:\mathbb{R}^2_+\to\mathbb{R}^2$ be a minimizing solution of* (1.1) *satisfying* (1.5)–(1.8)*. For any sequence $r_j\to\infty$, there exists a subsequence, still denoted by $\{r_j\}$, and a function $\tilde{u}\in \mathrm{BV}_{loc}(\mathbb{R}^2_+;\{a_1,...,a_N\})$ such that*

$$u_{r_j}(z) \to \tilde{u}(z), \quad \text{in } L^1_{loc}(\mathbb{R}^2_+;\mathbb{R}^2). \tag{2.21}$$

Here $u_{r_j}(z) := u(r_j z)$ and $\tilde{u}$ is a minimizing partition map that locally minimizes $J_*(\cdot,\Omega)$ in the sense of Definition 1.2.

Regarding the Radon measures induced by the energy density

$$\mu_j := \left(\frac{1}{2r_j}|\nabla u_{r_j}|^2 + r_j W(u_{r_j})\right)\mathcal{L}^2\llcorner\mathbb{R}^2_+,$$

it holds that

$$\mu_j \overset{*}{\rightharpoonup} \mu_{\tilde{u}} \quad \text{locally on } \overline{\mathbb{R}^2_+}, \tag{2.22}$$

where $\mu_{\tilde{u}}$ is the energy measure associated with J_*, i.e.

$$\begin{aligned}\mu_{\tilde{u}} := &\sum_{1\le i<j\le N} \sigma_{ij}\mathcal{H}^1 \llcorner (\partial^*\tilde{u}^{-1}(a_i)\cap\partial^*\tilde{u}^{-1}(a_j)\cap\mathbb{R}^2_+)\\ &+ d(\operatorname{tr}\tilde{u},a_1)\mathcal{H}^1\llcorner\{(x,0):x>0\} + d(\operatorname{tr}\tilde{u},a_2)\mathcal{H}^1\llcorner\{(x,0):x<0\}.\end{aligned} \tag{2.23}$$

Consequently, for every bounded open set $A\subset\mathbb{R}^2$ satisfying

$$\mu_{\tilde{u}}(\partial A\cap\overline{\mathbb{R}^2_+}) = 0,$$

we have

$$\lim_{j\to\infty} J_{r_j}(u_{r_j}, A\cap\mathbb{R}^2_+) = \lim_{j\to\infty}\frac{1}{r_j}J(u,(r_jA)\cap\mathbb{R}^2_+) = \mu_{\tilde{u}}(A\cap\overline{\mathbb{R}^2_+}). \tag{2.24}$$

Proof. We follow closely the compactness result of Sandier and Sternberg [35, Proposition 3.1], using the Γ-convergence techniques in Baldo [7] and the treatment of Dirichlet boundary condition in Gazoulis [24].

Write

$$u_j := u_{r_j}, \qquad g_j(x) := u_0(r_j\,x), \qquad g(x) := a_1\chi_{\{x>0\}} + a_2\chi_{\{x<0\}}.$$

Lemma 2.5 implies that

$$J_{r_j}(u_j, \overline{B^+_\rho}) \le C\rho, \qquad \forall\rho>0.$$

Therefore, up to a possible subsequence, μ_j converges weakly to a Radon measure μ on $\overline{\mathbb{R}^2_+}$. We extend u_j below the x-axis (still denoted by u_j) by defining

$$u_j(x,y) = g_j(x), \quad x\in\mathbb{R},\ y\in(-1,0).$$

For every $\delta\in(-1,0)$,

$$J_{r_j}(u_j,\mathbb{R}\times[\delta,0]) = |\delta|\cdot\int_{\mathbb{R}}\frac{1}{2}|u_0'|^2 + W(u_0)\,dx = C_0|\delta|,$$

where $C_0 = \int_{\mathbb{R}}\frac{1}{2}|u_0'|^2+W(u_0)\,dx$. Using the classic compactness theorem for multiwell Allen-Cahn systems, see for example [7, Proposition 4.1], together with the uniform L^∞ bound for u_j, we have

$$u_j\to\tilde{u} \quad \text{in } L^1_{loc}(\{(x,y):y>-1\};\mathbb{R}^2),$$

for some $\tilde{u}\in \mathrm{BV}_{loc}(\{(x,y):y>-1\};\{a_1,...,a_N\})$. Since $\|g_j-g\|_{L^2(\mathbb{R};\mathbb{R}^2)}\to 0$, it holds that

$$\tilde{u}(x,y) = g(x), \quad a.e.\text{ on } \{(x,y):-1<y<0\}.$$

For the extended energy density measure

$$\mu_j^* := \left(\frac{1}{2r_j}|\nabla u_{r_j}|^2 + r_jW(u_{r_j})\right)\mathcal{L}^2\llcorner\{(x,y):y>-1\},$$

we have

$$\mu_j^* \overset{*}{\rightharpoonup} \mu + C_0\mathcal{H}^1\llcorner\{(0,y):-1<y<0\}.$$

By the lower bound part of Γ-convergence, see [7, (2.8)], we have for any $\delta\in(-1,0)$

$$\mu + C_0\mathcal{H}^1\llcorner\{(0,y):\delta<y<0\} \ge \sum_{1\le i<j\le N}\sigma_{ij}\mathcal{H}^1\llcorner(\partial^*\tilde{u}^{-1}(a_i)\cap\partial^*\tilde{u}^{-1}(a_j)\cap\{y>\delta\}).$$

Taking $\delta\to 0-$, we get

$$\mu\ge\mu_{\tilde{u}} \quad \text{on } \overline{\mathbb{R}^2_+}, \tag{2.25}$$

where $\mu_{\tilde{u}}$ is defined by (2.23). Here the interface mismatch between the interior trace $\mathrm{tr}_{\partial\mathbb{R}^2_+}\tilde{u}$ and the exterior phase g produces the boundary term in (2.23).

For the upper bound, we can invoke the construction in [24, proof of Theorem 1.1] to obtain, for every $t>0$ satisfying

$$\mu_{\tilde{u}}(\partial B_t \cap \overline{\mathbb{R}^2_+}) = 0,$$

a sequence of maps

$$w_j \in H^1(B_t^+;\mathbb{R}^2)\cap L^\infty(B_t^+;\mathbb{R}^2), \quad \sup_j \|w_j\|_{L^\infty(B_t^+)} < \infty,$$

such that

$$\begin{aligned} \mathrm{tr}_{\{(x,0):-t<x<t\}} w_j = g_j, \qquad w_j \to \tilde{u} \quad \text{in } L^1(B_t^+;\mathbb{R}^2),\\ \limsup_{j\to\infty} J_{r_j}(w_j, B_t^+) \le \mu_{\tilde{u}}(B_t \cap \overline{\mathbb{R}^2_+}). \end{aligned}$$

Fix $s<t$ such that $\mu(\partial B_s \cap \overline{\mathbb{R}^2_+}) = 0$. Since u_j and w_j both converge in L^1 (can be strengthened to L^2 due to uniform boundedness) to $\tilde{u}$ in the half-annulus $B_t^+ \setminus B_s^+$, using the gluing estimate in [24, Lemma 2.1] one can construct a competitor in B_t^+, which agrees with u_j on $\partial B_t \cap \mathbb{R}^2_+$ and w_j on $\partial B_s \cap \mathbb{R}^2_+$, and obtain

$$J_{r_j}(u_j, B_s^+) \le J_{r_j}(w_j, B_t^+) + o(1).$$

Letting $j\to\infty$ and then $t\downarrow s$ gives

$$\mu(B_s \cap \overline{\mathbb{R}^2_+}) \le \mu_{\tilde{u}}(B_s \cap \overline{\mathbb{R}^2_+}), \quad \forall s \text{ satisfying } \mu(\partial B_s \cap \overline{\mathbb{R}^2_+}) = 0,$$

which together with (2.25) gives (2.22). The minimality of $\tilde{u}$ follows from a similar recovery sequence construction argument. For a competitor partition w agreeing with $\tilde{u}$ outside a compact set, apply the same recovery construction to w. Choose the gluing annulus outside the compact set, where the recovery sequence w_j and u_j both converge to $\tilde{u}$. The minimality of u_j, the gluing estimate, and energy density measure convergence then give the minimizing property of $\tilde{u}$. This completes the proof.

□

3. Classification of blow-down limits

Throughout this section, we assume (H_1), (H_2), and either (H_3) or (H_3'). Let u be a minimizing solution in the sense of Definition 1.1 satisfying (1.5)–(1.8). Let $\tilde{u} \in \mathrm{BV}_{loc}(\mathbb{R}^2_+;\{a_1,...,a_N\})$ be a sequential L^1_{loc} blow-down limit of $u_{r_j}(z)$ obtained in Proposition 2.6. In particular, $\tilde{u}$ is a minimizing partition map in the sense of Definition 1.2, and the energy measure convergence (2.22) holds.

Define the energy density of $\tilde{u}$ normalized by the radius r:

$$\Theta(r) := \frac{J_*(\tilde{u}, B_r \cap \overline{\mathbb{R}^2_+})}{r}.$$

Since $\tilde{u} \in \mathrm{BV}_{loc}(\mathbb{R}^2_+;\{a_1,...,a_N\})$, for almost every $r>0$, the angular slice

$$h_r(\theta) := \tilde{u}(r\cos\theta, r\sin\theta) \in \mathrm{BV}((0,\pi);\{a_1,...,a_N\}).$$

Let $\mathfrak{J}_r$ denote the discontinuity set of h_r in $(0,\pi)$:

$$\mathfrak{J}_r := \{\theta\in(0,\pi) : h_r(\theta-) \neq h_r(\theta+)\}.$$

Further define

$$Q(r) := \sum_{\theta\in\mathfrak{J}_r} d(h_r(\theta-), h_r(\theta+)) + d(h_r(0+), a_1) + d(h_r(\pi-), a_2), \tag{3.26}$$

where the last two terms count the possible mismatches with the prescribed phase $g = a_1\chi_{\mathbb{R}_+} + a_2\chi_{\mathbb{R}_-}$ on $\partial\mathbb{R}^2_+$. Finally denote by

$$\gamma_{ij} := \partial^*\tilde{u}^{-1}(a_i) \cap \partial^*\tilde{u}^{-1}(a_j) \cap \mathbb{R}^2_+$$

the interior interface between a_i and a_j phases.

The following lemma asserts the monotonicity for $\Theta(r)$.

Lemma 3.1. *For the minimizing partition $\tilde{u}$, it holds that for almost every $r > 0$,*

$$\Theta(r) \le Q(r). \tag{3.27}$$

Moreover,

$$(r\Theta(r))' \ge Q(r), \quad \text{in the distributional sense.} \tag{3.28}$$

Consequently, $\Theta(r)$ has a nondecreasing representative. If $\Theta(r)$ is constant on $(0,\infty)$, then $\tilde{u}$ is homogeneous, i.e. $\tilde{u}$ admits a representative depending only on θ.

Remark 3.1. The statement of Lemma 3.1 applies to any minimizing partition v in the sense of Definition 1.2. Since our objective is to classify the blow-down limit $\tilde{u}$, we state the lemma and give the proof for $\tilde{u}$.

Proof. The lemma is a standard cone comparison monotonicity formula, see for example [45, Theorem 8.2] in the setting of minimizing flat chains. We present a short proof to account for the boundary mismatch in the half-plane setting.

For almost every r, $h_r(\theta)$ is a BV function with finite jump set $\mathfrak{J}_r$ and satisfies

$$h_r(\theta) = \mathrm{tr}_{\partial B_r\cap\mathbb{R}^2_+}\tilde{u}, \quad h_r(0+) = \mathrm{tr}\,\tilde{u}(r), \quad h_r(\pi-) = \mathrm{tr}\,\tilde{u}(-r). \tag{3.29}$$

For such r, we consider the homogeneous extension of h_r inside B_r^+

$$w(t,\theta) = h_r(\theta), \quad \forall t\in(0,r),\ \theta\in(0,\pi).$$

Then w is a competitor for $\tilde{u}$ in B_r^+, which by minimality of $\tilde{u}$ implies

$$J_*(\tilde{u}, B_r\cap\overline{\mathbb{R}^2_+}) \le rQ(r),$$

which establishes (3.27).

We next set τ_{ij} as the unit tangent vector field along γ_{ij}. By coarea formula, for any test function $\eta\in C_C^\infty((0,\infty);[0,\infty))$,

$$\begin{aligned}\int_0^\infty Q(r)\eta(r)\,dr &= \sum_{i<j}\sigma_{ij}\int_{\gamma_{ij}}\left|\tau_{ij}\cdot\frac{z}{|z|}\right|\eta(|z|)\,d\mathcal{H}^1 + \int_0^\infty d(\mathrm{tr}\,\tilde{u}, a_1)\eta(x)\,dx\\ &\quad + \int_{-\infty}^0 d(\mathrm{tr}\,\tilde{u}, a_2)\eta(|x|)\,dx\\ &\le \int_{\overline{\mathbb{R}^2_+}}\eta(|z|)\,d\mu_{\tilde{u}}(z) = \int_0^\infty \eta(r)d(r\Theta(r)),\end{aligned} \tag{3.30}$$

which implies (3.28). Combining (3.27) and (3.28) yields

$$\Theta(r)' = \frac{(r\Theta(r))'}{r} - \frac{\Theta(r)}{r} \ge \frac{Q(r)-\Theta(r)}{r} \ge 0, \qquad \text{distributionally.}$$

If $\Theta(r) \equiv c$ on $(0,\infty)$, then (3.27) and (3.28) imply that $Q(r) \equiv c$. In particular, (3.30) takes equality for every test function η, which implies

$$\left|\tau_{ij} \cdot \frac{z}{|z|}\right| = 1, \qquad \mathcal{H}^1\text{-a.e. on } \gamma_{ij},\ \forall i<j.$$

This further implies that $r \mapsto \tilde{u}(r,\theta)$ is constant for almost every $\theta \in (0,\pi)$, which proves the homogeneity of $\tilde{u}$.

□

Proposition 3.2. *There exists $\alpha \geq \sigma_{12}$ such that*

$$\lim_{r\to\infty} \frac{J(u, B_r^+)}{r} = \alpha, \tag{3.31}$$

$$\mu_{\tilde{u}}(B_r \cap \overline{\mathbb{R}^2_+}) = \alpha r \quad \textit{for every } r \in (0,\infty). \tag{3.32}$$

Consequently, every blow-down limit $\tilde{u}$ is homogeneous.

Proof. Let C_0 be the constant in Lemma 2.5 such that $J(u, B_r^+) \leq C_0 r$. For any $t \in [\sigma_{12}, C_0]$, suppose there exists a radius r such that $h_r(\theta)$ is a BV function satisfying (3.29) and

$$Q(r) = t.$$

Then t belongs to the discrete set

$$\mathfrak{S} := \{t \in [\sigma_{12}, C_0] : \exists \{n_{ij}\}_{i<j} \subset \mathbb{N}_0 = \{0,1,2,...\} \ \text{ s.t. } \sum_{i<j} \sigma_{ij}\, n_{ij} = t\},$$

since every term in the sum of $Q(r)$ is one of $\{\sigma_{ij},\ 1 \leq i < j \leq N\}$. Here $Q(r) \geq \sigma_{12}$ follows from the triangle inequality and the boundary terms in (3.26).

Assume by contradiction that the limit $\lim_{r\to\infty} \frac{J(u,B_r^+)}{r}$ does not exist. Choose two numbers α_1, α_2 such that

$$\liminf_{r\to\infty} \frac{J(u, B_r^+)}{r} < \alpha_1 < \alpha_2 < \limsup_{r\to\infty} \frac{J(u, B_r^+)}{r}, \qquad \alpha_2 \notin \mathfrak{S}.$$

Then by continuity of $J(u, B_r^+)$, one can find two sequences $r_j \to \infty$, $R_j \to \infty$, such that

$$r_j < R_j, \quad \frac{J(u, B_r^+)}{r} \leq \alpha_2 \ \text{ for } r \in [r_j, R_j],$$
$$\frac{J(u, B_{r_j}^+)}{r_j} = \alpha_2, \quad \frac{J(u, B_{R_j}^+)}{R_j} = \alpha_1.$$

Therefore we have

$$\alpha_2 r_j \leq \alpha_1 R_j \quad \Rightarrow \quad \frac{R_j}{r_j} \geq \frac{\alpha_2}{\alpha_1} =: \kappa.$$

Using Proposition 2.6, let $\tilde{u}$ be a sequentially blow-down limit of u_{r_j} after passing to a subsequence, still denoted by u_{r_j}. For any $r \in (1,\kappa)$ such that $\mu_{\tilde{u}}(\partial B_r \cap \mathbb{R}^2_+) = 0$, it holds that

$$r\Theta(r) = \lim_{j\to\infty} \frac{J(u, B^+_{r_j r})}{r_j} \in [\alpha_2, \alpha_2 r],$$

which gives $\Theta(r) \leq \alpha_2$ for almost every $r \in (1,\kappa)$. On the other hand, monotonicity of $\Theta(r)$ implies

$$\Theta(s) \geq \Theta(r) \geq \frac{\alpha_2}{r}, \quad \text{ for } 1 < r < s < \kappa.$$

Taking the continuity radii $r \downarrow 1$ yields $\Theta(s) \geq \alpha_2$ for almost every $s \in (1,\kappa)$. Hence,

$$\Theta(r) \equiv \alpha_2, \quad r \in (1,\kappa).$$

Lemma 3.1 then implies $Q(r) \equiv \alpha_2$ almost everywhere on $(1, \kappa)$, contradicting $\alpha_2 \notin \mathfrak{S}$. Therefore $\frac{J(u,B_r^+)}{r}$ must converge to some $\alpha > 0$.

For any blow-down limit $\tilde{u}$ and almost every $r > 0$, we have

$$\Theta(r) = \frac{J_*(\tilde{u}, B_r \cap \overline{\mathbb{R}^2_+})}{r} = \lim_{j\to\infty} \frac{J(u, B^+_{r_j r})}{r_j r} = \alpha.$$

The homogeneity of $\tilde{u}$ follows immediately from Lemma 3.1. Its angular representative $\tilde{u}(\theta) = \tilde{u}(r, \theta)$ is a BV function on $(0, \pi)$ with finitely many jumps. (3.32) is a direct consequence of $Q(r) \equiv \Theta(r) \equiv \alpha$ for a.e. r. Finally, $\alpha \geq \sigma_{12}$ thanks to the prescribed boundary data.

□

By Proposition 3.2, we can write

$$\tilde{u}(r, \theta) = h(\theta) \in \mathrm{BV}((0, \pi); \{a_1, ..., a_N\}), \quad r \in (0, \infty), \quad \theta \in (0, \pi).$$

Set

$$\mathfrak{J} := \{\theta \in (0, \pi) : h(\theta-) \neq h(\theta+)\},$$

as the set of jump angles. For the minimizing solution u, we define the quantities

$$\mathcal{W}(r) := \frac{1}{r} \int_{B_r^+} W(u)\, dz,$$
$$\mathcal{N}(r) := \frac{1}{r} \int_{B_r^+} \frac{1}{2} |\nabla u|^2\, dz.$$

The following lemma shows an asymptotic energy equipartition property for u at large scales.

Lemma 3.3. *Let α be the constant in* (3.31)*. Then*

$$\lim_{r\to\infty} \mathcal{W}(r) = \lim_{r\to\infty} \mathcal{N}(r) = \frac{\alpha}{2}. \tag{3.33}$$

Moreover,

$$\lim_{r\to\infty} \frac{1}{r} \int_{B_r^+} \left(\sqrt{W(u)} - \frac{|\nabla u|}{\sqrt{2}} \right)^2 dz = 0, \tag{3.34}$$

and for every $0 < \lambda_1 < \lambda_2$,

$$\lim_{r\to\infty} \int_{A^+_{\lambda_1 r, \lambda_2 r}} \frac{1}{|z|} |\partial_r u|^2\, dz = 0. \tag{3.35}$$

Proof. Fix $\lambda_2 > \lambda_1 > 0$. Recall that $\tilde{u}$ is the L^1_{loc} limit of u_{r_j} for a sequence $r_j \to \infty$ and satisfies

$$\tilde{u}(r, \theta) = h(\theta),$$
$$\lim_{r\to\infty} \frac{J(u, B_r^+)}{r} = \alpha = \sum_{\theta\in\mathfrak{J}} d(h(\theta-), h(\theta+)) + d(h(0+), a_1) + d(h(\pi-), a_2).$$

It follows that

$$E_j := \int_{A^+_{\lambda_1,\lambda_2}} \left(\frac{1}{2r_j} |\nabla u_{r_j}|^2 + r_j W(u_{r_j}) \right) dz = \frac{J(u, B^+_{\lambda_2 r_j}) - J(u, B^+_{\lambda_1 r_j})}{r_j} \longrightarrow \alpha(\lambda_2 - \lambda_1). \tag{3.36}$$

Set

$$F_j := \int_{A^+_{\lambda_1,\lambda_2}} \sqrt{2W(u_{r_j})} \frac{|\partial_\theta u_{r_j}|}{|z|}\, dz.$$

It is clear that $F_j \le E_j$. Fubini's theorem implies, for almost every $r \in (\lambda_1, \lambda_2)$, $u_{r_j}(r,\cdot)$ (after extracting a possible subsequence) converges to $h(\theta)$ in $L^1(0,\pi)$, which further implies for such r,

$$\liminf_{j\to\infty} \int_0^\pi \sqrt{2W(u_{r_j}(r,\theta))}|\partial_\theta u_{r_j}(r,\theta)|\,d\theta \ge \alpha,$$

since the left-hand side has to count the energy cost for all phase transitions at jump points of h and possible boundary mismatches. This together with (3.36) gives

$$0 \le \lim_{j\to\infty}(E_j - F_j) \le \alpha(\lambda_2-\lambda_1) - \alpha(\lambda_2-\lambda_1) = 0.$$

Therefore,

$$\begin{aligned} 0 &= \lim_{j\to\infty} E_j - F_j \\ &= \lim_{j\to\infty} \int_{A^+_{\lambda_1,\lambda_2}} \frac{1}{2r_j}|\partial_r u_{r_j}|^2\,dz + \lim_{j\to\infty}\int_{A^+_{\lambda_1,\lambda_2}} \left|\frac{1}{\sqrt{2r_j}}\frac{|\partial_\theta u_{r_j}|}{|z|} - \sqrt{r_j W(u_{r_j})}\right|^2 dz. \end{aligned} \tag{3.37}$$

Furthermore,

$$\lim_{j\to\infty}\int_{A^+_{\lambda_1,\lambda_2}} \left|\frac{|\nabla u_{r_j}|}{\sqrt{2r_j}} - \sqrt{r_j W(u_{r_j})}\right|^2 dz \le \lim_{j\to\infty} E_j - F_j = 0. \tag{3.38}$$

By Cauchy-Schwarz, we have

$$\begin{aligned} &\left|\int_{A^+_{\lambda_1,\lambda_2}} \frac{|\nabla u_{r_j}|^2}{2r_j}\,dz - \int_{A^+_{\lambda_1,\lambda_2}} r_j W(u_{r_j})\,dz\right| \\ &\le \left|\int_{A^+_{\lambda_1,\lambda_2}} \left(\frac{|\nabla u_{r_j}|}{\sqrt{2r_j}} - \sqrt{r_j W(u_{r_j})}\right)\left(\frac{|\nabla u_{r_j}|}{\sqrt{2r_j}} + \sqrt{r_j W(u_{r_j})}\right) dz\right| \\ &\le \left(\int_{A^+_{\lambda_1,\lambda_2}} \left|\frac{|\nabla u_{r_j}|}{\sqrt{2r_j}} - \sqrt{r_j W(u_{r_j})}\right|^2 dz\right)^{\frac12} (2E_j)^{\frac12} \\ &\longrightarrow 0, \quad \text{as } j\to\infty. \end{aligned} \tag{3.39}$$

Rescaling (3.37), (3.38) and (3.39) back to the original variables, we obtain

$$\lim_{j\to\infty}\int_{A^+_{\lambda_1 r_j,\lambda_2 r_j}} \frac{|\partial_r u|^2}{|z|}\,dz = 0, \tag{3.40}$$

$$\lim_{j\to\infty}\frac{1}{r_j}\int_{A^+_{\lambda_1 r_j,\lambda_2 r_j}} \frac{|\nabla u|^2}{2}\,dz = \lim_{j\to\infty}\frac{1}{r_j}\int_{A^+_{\lambda_1 r_j,\lambda_2 r_j}} W(u)\,dz = \frac{\lim_{j\to\infty} E_j}{2} = \frac{\alpha(\lambda_2-\lambda_1)}{2}, \tag{3.41}$$

$$\lim_{j\to\infty}\frac{1}{r_j}\int_{A^+_{\lambda_1 r_j,\lambda_2 r_j}} \left(\frac{|\nabla u|}{\sqrt{2}} - \sqrt{W}\right)^2 dz = 0. \tag{3.42}$$

(3.40) gives (3.35) along the sequence $r_j \to \infty$. In (3.41) and (3.42), we can take $\lambda_2 = 1$ and let $\lambda_1 \downarrow 0$ to recover (3.33) and (3.34) along the same sequence $r_j \to \infty$. All conclusions still hold for $r \to \infty$, since for any $R_j \to \infty$, one can extract a subsequence satisfying the convergence results.

□

Next we give a classification of homogeneous minimizing partitions in $\mathbb{R}^+_2$. Suppose $v \in \mathrm{BV}_{loc}$ is a minimizing partition in the sense of Definition 1.2 and satisfies $v(\lambda z) = v(z)$ for almost every $z \in \mathbb{R}^2_+$, $\lambda \in \mathbb{R}_+$. Let $h_v(\theta) \in \mathrm{BV}((0,\pi);\{a_1,...,a_N\})$ be the angular representative of v satisfying

$$v(r,\theta) = h_v(\theta).$$

Under (H_3), we define the angles

$$\theta_i := \arccos\left(\frac{\sigma_{jk}^2 - \sigma_{ik}^2 - \sigma_{ij}^2}{2\sigma_{ij}\sigma_{ik}}\right), \quad \{i,j,k\} = \{1,2,3\}.$$

Then the triangle inequality (1.4) implies $\theta_i \in (0,\pi)$ for every i, and these angles are the supplements of the angles for a triangle with three side lengths σ_{12}, σ_{23}, σ_{13}. In particular they satisfy Young's Law (1.12)

Proposition 3.4. *Assume (H_3) or (H'_3). For any homogeneous minimizing partition v in Definition 1.2, its angular representative $h_v(\theta)$ must have one of the following forms.*

(1) For some $\alpha \in [0,\pi]$,

$$h_v(\theta) = \begin{cases} a_1, & 0 < \theta < \alpha, \\ a_2, & \alpha < \theta < \pi. \end{cases} \tag{3.43}$$

(2) For some $i \in \{3,...,N\}$ and angles $0 \le \alpha_1 < \alpha_2 \le \pi$,

$$h_v(\theta) = \begin{cases} a_1, & 0 < \theta < \alpha_1, \\ a_i, & \alpha_1 < \theta < \alpha_2, \\ a_2, & \alpha_2 < \theta < \pi. \end{cases} \tag{3.44}$$

Moreover, the angle of the third phase a_i satisfies

$$\alpha_2 - \alpha_1 \ge \begin{cases} \theta_3, & \text{under } (\mathrm{H}_3), \\ \frac{2\pi}{3}, & \text{under } (\mathrm{H}'_3). \end{cases}$$

In the above formula, empty angular intervals are omitted.

Conversely, each angular function h_v satisfying (1) or (2) defines a homogeneous minimizing partition. Thus, this gives a necessary and sufficient condition for homogeneous minimizing partitions.

Proof. Take a minimizing partition $v(r,\theta) = h_v(\theta)$. We denote the jump angles of h_v by

$$0 \le \alpha_1 < \alpha_2 < \cdots < \alpha_k \le \pi.$$

Here $\alpha_1 = 0$ if $h_v(0+) \ne a_1$, which means there is a boundary mismatch between the interior trace of v at $\{(x,0) : x > 0\}$ and a_1 phase. Similarly, the last jump $\alpha_k = \pi$ if $h_v(\pi-) \ne a_2$.

Let $i_0, i_1, ..., i_{k-1}, i_k$ be the indices of phases occurring counterclockwise:

$$h_v(\theta) = \begin{cases} a_1, & \theta \in (0,\alpha_1), \\ a_{i_j}, & \theta \in (\alpha_j,\alpha_{j+1}), \quad j = 1,...,k-1, \\ a_2, & \theta \in (\alpha_k,\pi). \end{cases}$$

Set the endpoint indices as $i_0 = 1$, $i_k = 2$. The ray $\{\theta = \alpha_j\}$ is an interface with weighted energy density $\sigma_{i_{j-1}i_j}$.

Consider two adjacent jump angles $\alpha_j < \alpha_{j+1}$. For convenience we write

$$a_{i_{j-1}} = a, \quad a_{i_j} = b, \quad a_{i_{j+1}} = c.$$

By definition, b is the phase between α_j and α_{j+1}, a is the previous phase on the right and c is the next phase on the left, see Figure 1.

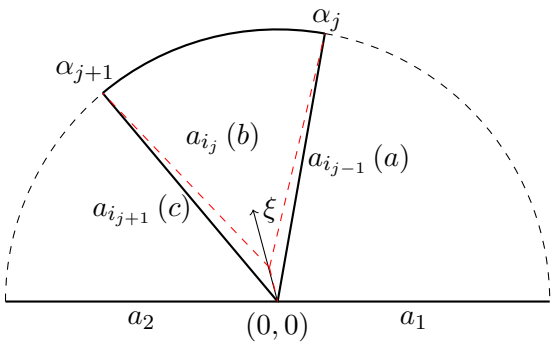

FIGURE 1. Each phase angle must be at least the corresponding Young angle.

For any angle $\alpha \in [0,\pi]$, set $\mathbf{e}_\alpha = (\cos\alpha, \sin\alpha)$. If $\alpha_{j+1} - \alpha_j = \pi$, then $j = 1$, $0 = \alpha_1 < \alpha_2 = \pi$, $b = a_i$ for some $i \in \{3, ..., N\}$, $h_v(\theta)$ has the form (3.44). Therefore, we only need to consider the case $\alpha_{j+1} - \alpha_j < \pi$. Define the direction

$$\xi = \sigma_{ab}\mathbf{e}_{\alpha_j} + \sigma_{bc}\mathbf{e}_{\alpha_{j+1}},$$

which lies strictly between $\mathbf{e}_{\alpha_j}$ and $\mathbf{e}_{\alpha_{j+1}}$. On the unit half-disk B_1^+, denote the sector of b-phase by

$$S := \{(r\cos\theta, r\sin\theta) : r \in (0,1), \theta \in (\alpha_j, \alpha_{j+1})\}.$$

We next perturb the interface $\{r\mathbf{e}_{\alpha_j} : r \in (0,1)\} \cup \{r\mathbf{e}_{\alpha_{j+1}} : r \in (0,1)\}$ in S by pinching together a-phase and c-phase along the ξ direction, thus creating a triple junction of a, b, c inside S. More precisely, for $0 < t \ll 1$, we replace $\{r\mathbf{e}_{\alpha_j} : r \in (0,1)\} \cup \{r\mathbf{e}_{\alpha_{j+1}} : r \in (0,1)\}$ by the union of segments joining $t\frac{\xi}{|\xi|}$ to $\mathbf{e}_{\alpha_\mathbf{j}}$ and $\mathbf{e}_{\alpha_{j+1}}$, and a length-$t$ a-c interface joining $t\frac{\xi}{|\xi|}$ to the origin. In case $a = c$, omit the last small a-c interface. We compute the energy change on B_1^+

$$\begin{aligned}
\Delta J_* &= \left(\left|t\frac{\xi}{|\xi|} - \mathbf{e}_{\alpha_\mathbf{j}}\right| - 1\right)\sigma_{ab} + \left(\left|t\frac{\xi}{|\xi|} - \mathbf{e}_{\alpha_{\mathbf{j+1}}}\right| - 1\right)\sigma_{bc} + t\sigma_{ac} \\
&= \frac{-t}{|\xi|}\xi \cdot (\sigma_{ab}\mathbf{e}_{\alpha_j} + \sigma_{bc}\mathbf{e}_{\alpha_{j+1}}) + t\sigma_{ac} + O(t^2) \\
&= t(\sigma_{ac} - |\xi|) + O(t^2).
\end{aligned}$$

By minimality we have

$$\sigma_{ac} \geq |\sigma_{ab}\mathbf{e}_{\alpha_j} + \sigma_{bc}\mathbf{e}_{\alpha_{j+1}}| \;\Rightarrow\; \sigma_{ac}^2 \geq \sigma_{ab}^2 + \sigma_{bc}^2 + 2\cos(\alpha_{j+1} - \alpha_j)\sigma_{ab}\sigma_{bc},$$

Since cosine function is decreasing on $(0,\pi)$, we have

$$\alpha_{j+1} - \alpha_j \geq \theta_b = \arccos\left(\frac{\sigma_{ac}^2 - \sigma_{ab}^2 - \sigma_{bc}^2}{2\sigma_{ab}\sigma_{bc}}\right), \tag{3.45}$$

where θ_b is the angle of the b-phase at a triple junction of phases a, b, c determined by (1.12). This also shows $a \neq c$ when $\alpha_{j+1} - \alpha_j < \pi$.

Under (H_3), i.e. $N = 3$, we first claim that $k \leq 2$. Otherwise assume $k \geq 3$, by the pinching argument, there exist $i, j \in \{1, 2, 3\}$, $i \neq j$, such that

$$\alpha_3 - \alpha_1 = (\alpha_3 - \alpha_2) + (\alpha_2 - \alpha_1) \geq \theta_i + \theta_j > \pi,$$

which yields a contradiction. When $k = 2$, the middle phase must be a_3, which verifies the form (3.44) with the angle condition $\alpha_2 - \alpha_1 \geq \theta_3$. When $k = 1$, there is only one a_1-a_2 phase transition, which corresponds to (3.43).

Under (H_3'), we also assert that $k \leq 2$, since otherwise there will be at least two middle phases, each with an opening angle greater than or equal to $\frac{2\pi}{3}$. Then $k = 2$ gives (3.44) with the third phase index $i \in \{3, ..., N\}$ and $k = 1$ gives the form (3.43).

Next we show that the partitions in (1) and (2) are minimizing. First we consider the 1-jump case (3.43). Let $K \subset \mathbb{R}^2_+$ be a bounded Lipschitz domain and $\nu_\alpha := (-\sin\alpha, \cos\alpha)$. Consider a competitor $w = \sum_i a_i \chi_{E_i}$ agreeing with v outside K and on $\partial K \cap \mathbb{R}^2_+$ in the sense of trace. Thus $\mathrm{tr}_{\partial K \cap \mathbb{R}^2_+} w$ only takes values a_1 and a_2. Denote by ν_{ij} the unit normal on $\partial^* E_i \cap \partial^* E_j$ pointing from E_i to E_j and set

$$\xi_1 := 0, \quad \xi_2 := -\sigma_{12}\nu_\alpha,$$
$$\begin{cases} \xi_3 := -\frac{\sigma_{13}\sigma_{12}}{\sigma_{13}+\sigma_{23}}\nu_\alpha, & \text{under } (\mathrm{H}_3), \\ \xi_i := -\frac{\sigma_{12}}{2}\nu_\alpha, \ \forall\, i \geq 3, & \text{under } (\mathrm{H}_3'). \end{cases}$$

Then it is straightforward to verify that

$$|\xi_1 - \xi_2| = \sigma_{12}, \qquad |\xi_i - \xi_j| < \sigma_{ij} \ \text{ for any } \{i, j\} \neq \{1, 2\}.$$

Using the divergence theorem we have

$$\begin{aligned} J_*(w, K) &\geq \sum_{i<j} \int_{K \cap \partial^* E_i \cap \partial^* E_j} (\xi_i - \xi_j) \cdot \nu_{ij} \, d\mathcal{H}^1 \\ &\quad + \int_{\partial K \cap \partial \mathbb{R}^2_+ \cap \{x>0\}} (\xi_{\mathrm{tr}w} - \xi_1) \cdot \nu_{\partial \mathbb{R}^2_+} \, d\mathcal{H}^1 + \int_{\partial K \cap \partial \mathbb{R}^2_+ \cap \{x<0\}} (\xi_{\mathrm{tr}w} - \xi_2) \cdot \nu_{\partial \mathbb{R}^2_+} \, d\mathcal{H}^1 \\ &= -\int_{\partial K} \xi_{\mathrm{tr}\, w} \cdot \nu_K \, d\mathcal{H}^1 + \int_{\partial K \cap \partial \mathbb{R}^2_+} (\xi_{\mathrm{tr}\, w} - \xi_g) \cdot \nu_{\partial \mathbb{R}^2_+} \, d\mathcal{H}^1 \\ &= -\int_{\partial K \cap \mathbb{R}^2_+} \xi_{\mathrm{tr}\, w} \cdot \nu_K \, d\mathcal{H}^1 - \int_{\partial K \cap \mathbb{R}^2_+} \xi_g \cdot \nu_{\partial \mathbb{R}^2_+} \, d\mathcal{H}^1. \end{aligned} \tag{3.46}$$

The integrals on the last line only depends on the trace on $\partial K \cap \mathbb{R}^2_+$ and the boundary condition $g = a_1\chi_{x>0} + a_2\chi_{x<0}$. For the partition v of the form (3.43), since $(\xi_1 - \xi_2) \cdot \nu_{12} = \sigma_{12}$ and there are only two phases a_1, a_2, the inequality becomes equality in (3.46) for v, which implies the minimality.

For the two-jump case (2) we follow the same idea. Under (H_3), set

$$\xi_3 = 0, \quad \xi_1 = \sigma_{13}\nu_{\alpha_1}, \quad \xi_2 = -\sigma_{23}\nu_{\alpha_2}. \tag{3.47}$$

One immediately has

$$|\xi_1 - \xi_3| = (\xi_1 - \xi_3) \cdot \nu_{13} = \sigma_{13}, \quad |\xi_2 - \xi_3| = (\xi_2 - \xi_3) \cdot \nu_{23} = \sigma_{23}.$$

Moreover,

$$|\xi_1 - \xi_2| = \sqrt{\sigma_{13}^2 + \sigma_{23}^2 + 2\sigma_{13}\sigma_{23}\cos(\alpha_2 - \alpha_1)} \leq \sqrt{\sigma_{13}^2 + \sigma_{23}^2 + 2\sigma_{13}\sigma_{23}\cos(\theta_3)} = \sigma_{12}.$$

Thus the same computation as in (3.46) can be carried out for this new set of $\{\xi_i\}$ and equality holds for v of the form (3.44), which proves the minimality. Under (H_3'), simply replace the index 3 in (3.47) by the middle phase index i. For any unused index j, just assign $\xi_j = 0$. Then the proof still works.

□

Corollary 3.5. *By Proposition 3.2 and 3.4, for the minimizing solution u, it holds that*

$$\lim_{r\to\infty}\frac{J(u,B_r^+)}{r}=\begin{cases}\sigma_{12}, & \text{some blow-down has the form (3.43)},\\ \sigma_{13}+\sigma_{23}, & \text{under } (\mathrm{H}_3)\text{, some blow-down has the form (3.44)},\\ 2\sigma, & \text{under } (\mathrm{H}_3')\text{, some blow-down has the form (3.44)}.\end{cases}$$

In particular, all sequential blow-downs belong to the same family, either the one-jump case (1) or the two-jump case (2). In addition, under (H_3'), the third phase a_i $(i\neq 1,2)$ for the two-jump solution remains the same for all blow-downs.

Proof. The classification of possible energy densities follows directly from Proposition 3.4. It suffices to prove the last statement: the third phase stays the same for all possible sequential blow-downs, given a blow-down limit has the form (3.44) under (H_3').

Since $\alpha_2-\alpha_1\geq\frac{2\pi}{3}$, we have for any $r_j\to\infty$, after passage to a subsequence, there exists $i\in\{3,...,N\}$ such that

$$\|u_{r_j}-a_i\|_{L^1(K)}\ \to\ 0,$$

where K denotes the sector

$$K:=\{(r\cos\theta,r\sin\theta):r\in(0,1),\ \theta\in(\frac{\pi}{3},\frac{2\pi}{3})\}.$$

Consequently,

$$\lim_{r\to\infty}\operatorname{dist}(\mathfrak{u}(r),\{a_3,...,a_N\})=0,$$

where $\mathfrak{u}(r):=\fint_K u_r(z)\,dz$. Note that $\mathfrak{u}(r)$ is continuous in r. Hence, one can identify a unique $i\in\{3,...,N\}$ such that $\lim_{r\to\infty}\mathfrak{u}(r)=a_i$, which implies the uniqueness of the third phase for all sequential blow-downs.

□

4. Uniqueness of the blow-down limit

In this section we show that the blow-down limit $\tilde{u}$ is independent of the choice of sequence $\{r_j\}$. The proof applies to both (H_3) and (H_3'). We will utilize the convergence of energy density in Proposition 2.6 and the classification of blow-down limits in Section 3.

Define the stress-energy tensor

$$T_{ij}:=\partial_i u\cdot\partial_j u-\delta_{ij}\left(\frac{1}{2}|\nabla u|^2+W(u)\right),\quad i,j\in\{1,2\}.$$

Then $\operatorname{div}(T_i)=\partial_j T_{ij}=\partial_i u\cdot(\Delta u-\nabla_u W(u))=0$ by (1.1). We also define the boundary flux on the semicircle $\partial B_r\cap\mathbb{R}^2_+$ as

$$\mathcal{F}(r):=-\int_{\partial B_r\cap\mathbb{R}^2_+}T\,\nu\,d\mathcal{H}^1,$$

where ν is the outer unit normal on $\partial B_r\cap\mathbb{R}^2_+$.

Lemma 4.1. *The inward normal derivative $\partial_y u(\cdot,0)$, which we abbreviate as $\partial_y u$ below, belongs to $L^2(\mathbb{R};\mathbb{R}^2)$. Moreover,*

$$\lim_{r\to+\infty}\mathcal{F}(r)=\left(-\int_{\mathbb{R}}u_0'\cdot\partial_y u\,dx,\ \int_{\mathbb{R}}(\frac{1}{2}|u_0'|^2-\frac{1}{2}|\partial_y u|^2+W(u_0))\,dx\right)=:\widetilde{\mathcal{F}}.\tag{4.48}$$

If a L^1 sequential blow-down limit $\tilde{u}$ has jump angle(s) $\alpha_l \in [0,\pi]$ (either one jump angle as in (3.43) *or two jump angles as in* (3.44)*), with associated weight σ_l, then*

$$\widetilde{\mathcal{F}} = \sum_l \sigma_l \mathbf{e}_{\alpha_l}. \tag{4.49}$$

Proof. Since $\operatorname{div}(T) = 0$, the divergence theorem implies

$$\mathcal{F}(r) = \int_{-r}^{r} T\,(0,-1)^T\,dx = \left(-\int_{-r}^{r} u_0' \cdot \partial_y u\,dx,\ \int_{-r}^{r} \big(\frac{1}{2}|u_0'|^2 - \frac{1}{2}|\partial_y u|^2 + W(u_0)\big)\,dx \right). \tag{4.50}$$

Let $\mathcal{F}_2(r)$ denote the second component of $\mathcal{F}(r)$. Since

$$\begin{aligned}
|\mathcal{F}_2(r)| &\le \int_{\partial B_r \cap \mathbb{R}^2_+} \sqrt{T_{21}^2 + T_{22}^2}\,d\mathcal{H}^1 \\
&\le \int_{\partial B_r \cap \mathbb{R}^2_+} \sqrt{(\frac{1}{2}|\nabla u|^2 + W(u))^2 - 2|\partial_y u|^2 W(u)}\,d\mathcal{H}^1 \\
&\le \int_{\partial B_r \cap \mathbb{R}^2_+} \left(\frac{1}{2}|\nabla u|^2 + W(u)\right) d\mathcal{H}^1 = \frac{d\,J(u, B_r^+)}{dr}, \quad \text{a.e. } r > 0.
\end{aligned}$$

By Lemma 2.5, $J(u, B_r^+) \le Cr$. One can find a sequence $R_k \to \infty$ such that

$$|\mathcal{F}_2(R_k)| \le \frac{d\,J(u, B_r^+)}{dr}\Big|_{r=R_k} \le 2C.$$

(4.50) and (1.6) then imply that

$$\int_{-R_k}^{R^k} \frac{1}{2}|\partial_y u(\cdot,0)|^2\,dx \le 2C + \int_{-R_k}^{R_k} \big(\frac{1}{2}|u_0'|^2 + W(u_0)\big)\,dx < \infty,$$

which further implies $\partial_y(u)(\cdot,0) \in L^2(\mathbb{R};\mathbb{R}^2)$. (4.48) follows immediately from (4.50) by taking $r \to \infty$.

To prove (4.49), we fix the sequence $r_j \to \infty$ such that $u_{r_j} \to \tilde{u}$ in L^1_{loc}. Take a cutoff function $\eta \in C_c^\infty((1,2);[0,\infty))$ such that $\int_1^2 \eta(r)\,dr = 1$. We have

$$\begin{aligned}
\int_1^2 \eta(r)\mathcal{F}(r_j r)\,dr &= \frac{1}{r_j}\int_{r_j}^{2r_j} \eta(\frac{r}{r_j}) \int_{\partial B_r \cap \mathbb{R}^2_+} \left(-(\nabla u)^T \partial_r u + (\frac{1}{2}|\nabla u|^2 + W(u))\nu \right) d\mathcal{H}^1 dr \\
&= \frac{1}{r_j}\int_{A^+_{r_j,2r_j}} \eta(\frac{|z|}{r_j}) \left(-(\nabla u)^T \partial_r u + (\frac{1}{2}|\nabla u|^2 + W(u))\nu \right) dz.
\end{aligned} \tag{4.51}$$

By (3.35) and Cauchy-Schwarz, the first term can be controlled by

$$\frac{\|\eta\|_{L^\infty}}{r_j} \left(\int_{A^+_{r_j,2r_j}} |\nabla u|^2\,dz \right)^{1/2} \left(\int_{A^+_{r_j,2r_j}} |\partial_r u|^2\,dz \right)^{1/2} \longrightarrow 0, \quad j \to \infty. \tag{4.52}$$

Moreover, rescaling and Proposition 2.6 (in particular, (2.22)) imply

$$\begin{aligned}
&\frac{1}{r_j}\int_{A^+_{r_j,2r_j}} \eta(\frac{|z|}{r_j})\left(\frac{1}{2}|\nabla u|^2+W(u)\right)\nu\,dz\\
&=\int_{A^+_{1,2}} \eta(|z|)\left(\frac{1}{2r_j}|\nabla u_{r_j}|^2+r_jW(u_{r_j})\right)\frac{z}{|z|}\,dz \qquad (4.53)\\
&\to\int_{\mathbb{R}^2_+}\eta(|z|)\frac{z}{|z|}\,d\mu_{\tilde{u}}=\int_{\mathbb{R}^2_+}\eta(|z|)\frac{z}{|z|}d\left(\sum_l \sigma_l\mathcal{H}^1\llcorner l_{\alpha_l}\right)=\sum_l\sigma_l\,\mathbf{e}_{\alpha_l}.
\end{aligned}$$

Note that the boundary mismatch has been counted, as the jump angle α_l could be 0 or π. Finally, we can derive (4.49) directly from (4.48) and (4.51)–(4.53).

□

Proposition 4.2. *There exists a unique blow-down limit $\tilde{u}$, with the structure characterized by Proposition 3.4, such that*

$$\lim_{r\to\infty}\|u_r-\tilde{u}\|_{L^1(K)}=0,$$

for any compact $K\subset\overline{\mathbb{R}^2_+}$.

Proof. By Corollary 3.5, it suffices to show that the jump angles are uniquely determined for all blow-downs. If there is only one jump, corresponding to the form (3.43), then (4.48) and (4.49) imply $\lim_{r\to\infty}\mathcal{F}(r)=\sigma_{12}\mathbf{e}_\alpha$, thus α is uniquely determined.

For the two-jump profile (3.44), $\widetilde{\mathcal{F}}=\sigma_{1i}\mathbf{e}_{\alpha_1}+\sigma_{i2}\mathbf{e}_{\alpha_2}$. Assume either ($H_3$) or ($H_3'$), and suppose there exists $(\tilde{\alpha}_1,\tilde{\alpha}_2)\neq(\alpha_1,\alpha_2)$ such that

$$\sigma_{1i}\mathbf{e}_{\alpha_1}+\sigma_{i2}\mathbf{e}_{\alpha_2}=\sigma_{1i}\mathbf{e}_{\tilde{\alpha}_1}+\sigma_{i2}\mathbf{e}_{\tilde{\alpha}_2}.$$

Then $\alpha_2-\alpha_1=\tilde{\alpha}_2-\tilde{\alpha}_1$ due to

$$|\widetilde{\mathcal{F}}|^2=\sigma_{1i}^2+\sigma_{i2}^2+2\sigma_{1i}\sigma_{i2}\cos(\alpha_2-\alpha_1)=\sigma_{1i}^2+\sigma_{i2}^2+2\sigma_{1i}\sigma_{i2}\cos(\tilde{\alpha}_2-\tilde{\alpha}_1)$$

Therefore, $\mathbf{e}_{\alpha_1}-\mathbf{e}_{\tilde{\alpha}_1}$ and $\mathbf{e}_{\alpha_2}-\mathbf{e}_{\tilde{\alpha}_2}$ are two parallel non-zero chords on B_1^+ with the same opening angle, which yields a contradiction.

□

Now we are ready to conclude the proof of Theorem 1.3.

Proof of Theorem 1.3. The statement in Theorem 1.3 follows immediately from Proposition 2.6, 3.4 and 4.2. □

5. Asymptotic behavior near the sharp interface

This section is devoted to the proof of Theorem 1.4, which establishes the asymptotic behavior of u near the interior sharp interface. Throughout this section we assume (H_1), (H_2), either (H_3) or (H_3'), and (H_4).

Let u be a minimizing solution of (1.1) satisfying (1.5)–(1.8), and let $\tilde{u}(r,\theta)=\tilde{u}(\theta)$ be its unique blow-down limit of u at infinity, i.e. for any bounded domain $\Omega\subset\mathbb{R}^2_+$,

$$\lim_{r\to\infty}\|u(r\cdot)-\tilde{u}\|_{L^1(\Omega)}=0.$$

In particular, we assume $\tilde{u}$ has the following structure (the intermediate phase is relabeled as a_3):

$$\tilde{u}(\theta)=\begin{cases}a_1, & \theta\in(0,\alpha_1)\\ a_3, & \theta\in(\alpha_1,\alpha_2),\\ a_2, & \theta\in(\alpha_2,\pi),\end{cases}\quad\text{for some } 0<\alpha_1<\alpha_2<\pi,\ \alpha_2-\alpha_1\geq\begin{cases}\theta_3, & (H_3)\\ \frac{2\pi}{3}, & (H_3').\end{cases}. \qquad (5.54)$$

The two discontinuity angles are α_1 and α_2, respectively. In the two-phase case where $\alpha_1 = \alpha_2 \in (0, \pi)$, the proof becomes simpler and is omitted.

We now set the angles

$$\varphi_1 := \frac{\alpha_1}{2}, \qquad \varphi_3 := \frac{\alpha_1 + \alpha_2}{2}, \qquad \varphi_2 := \frac{\alpha_2 + \pi}{2}, \tag{5.55}$$

where for each i, $\mathbf{e}_{\varphi_i}$ represents the direction of the bisector of a_i phase. Set $l_{\varphi_i} := \{r\mathbf{e}_{\varphi_i} : r \in (0, \infty)\}$ as the ray emanating from the origin and in direction φ_i. We first establish the following exponential decay estimate in the interior of phase regions away from the sharp interface.

Lemma 5.1. *Take an open domain $\Omega \subset \mathbb{R}^2_+$ satisfying $\tilde{u} \equiv a_i$ in Ω. For every $K \Subset \Omega$, there exist c, C, r_0, depending on W, u, K, such that for every $r > r_0$,*

$$\sup_{z \in K} \left(|u(rz) - a_i| + |\nabla u(rz)| + |\nabla^2 u(rz)| \right) \le Ce^{-cr}, \tag{5.56}$$

where ∇, ∇^2 mean derivatives of u evaluated at rz. In particular, on the rays l_{φ_i} for $i = 1, 2, 3$,

$$|u(r\mathbf{e}_{\varphi_i}) - a_i| + |\nabla u(r\mathbf{e}_{\varphi_i})| + |\nabla^2 u(r\mathbf{e}_{\varphi_i})| \le Ce^{-cr} \quad \text{for all sufficiently large } r. \tag{5.57}$$

Proof. Take $\rho > 0$ such that $\{z : \operatorname{dist}(z, K) \le \rho\} \subset \Omega$. Then we have

$$\fint_{B_{r\rho}(r\xi)} |u(z) - a_i| \, dz = \fint_{B_\rho(\xi)} |u_r(z) - a_i| \, dz \to 0 \text{ uniformly for } \xi \in K.$$

Applying Lemma 2.4 gives (5.56).

To show (5.57), simply choose Ω to be the a_i sector for $\tilde{u}$, and let K denote the disk $B_{r_i}(\mathbf{e}_{\varphi_i})$, where $r_i > 0$ such that $B_{2r_i}(\mathbf{e}_{\varphi_i}) \subset \Omega$. Then applying (5.56) to such choice of Ω and K gives (5.57). □

The following lemma provides energy lower bound and upper bound on B_R^+.

Lemma 5.2. *There is a constant $C = C(u, W)$ such that for every $R > 0$,*

$$(\sigma_{13} + \sigma_{23})R - C \le J(u, B_R^+) \le (\sigma_{13} + \sigma_{23})R + C. \tag{5.58}$$

Proof. Define

$$A(r) := \int_{\partial B_r \cap \mathbb{R}^2_+} \left(\frac{1}{2} |\partial_T u|^2 + W(u) \right) d\mathcal{H}^1.$$

We apply the estimate (5.57) on the rays l_{φ_i}, and Lemma 2.1 to obtain

$$\begin{aligned} J(u, B_R^+) &\ge \int_0^R A(r) \, dr \\ &\ge \int_0^R \left(\sigma_{13} + \sigma_{23} - C|u(r\mathbf{e}_{\varphi_1}) - a_1|^2 - C|u(r\mathbf{e}_{\varphi_3}) - a_3|^2 - C|u(r\mathbf{e}_{\varphi_2}) - a_2|^2 \right) dr \\ &\ge (\sigma_{13} + \sigma_{23})R - C \int_0^R Ce^{-2cr} \, dr \ge (\sigma_{13} + \sigma_{23})R - C(u, W), \end{aligned}$$

which yields the lower bound.

For the upper bound, we present a constructive proof. Fix $\gamma \le \delta_W$. Choose $0 < \delta < \gamma$ and $\varepsilon > 0$ to be chosen below. From (3.31) we have

$$\lim_{R \to \infty} \frac{1}{R} \int_R^{2R} A(r) \, dr = \sigma_{13} + \sigma_{32}.$$

Thus, for sufficiently large R, there exists $R_0 \in (R, 2R)$ such that

$$A(R_0) \le \sigma_{13} + \sigma_{32} + \varepsilon. \tag{5.59}$$

For such R_0, there exist

$$T_{13} := (\alpha_{13}^-, \alpha_{13}^+) \subset (\frac{3\alpha_1}{4}, \frac{3\alpha_1+\alpha_2}{4}), \qquad T_{32} := (\alpha_{32}^-, \alpha_{32}^+) \subset (\frac{3\alpha_2+\alpha_1}{4}, \frac{3\alpha_2+\pi}{4}),$$

satisfying

$$\begin{aligned} |u(R_0\mathbf{e}_\theta) - a_1| &\le \delta, \qquad \theta \in (\varphi_1, \alpha_{13}^-), \\ |u(R_0\mathbf{e}_\theta) - a_3| &\le \delta \qquad \theta \in (\alpha_{13}^+, \alpha_{32}^-), \\ |u(R_0\mathbf{e}_\theta) - a_2| &\le \delta \qquad \theta \in (\alpha_{32}^+, \varphi_2), \end{aligned}$$

thanks to the uniform convergence away from the sharp interface (5.56). We further define

$$\beta_{13}^- := \sup\{\theta \in \overline{T_{13}} : |u(R_0\mathbf{e}_\theta) - a_1| \le \delta\}, \qquad \beta_{13}^+ := \inf\{\theta \in \overline{T_{13}} : |u(R_0\mathbf{e}_\theta) - a_3| \le \delta\},$$
$$\beta_{32}^- := \sup\{\theta \in \overline{T_{32}} : |u(R_0\mathbf{e}_\theta) - a_3| \le \delta\}, \qquad \beta_{32}^+ := \inf\{\theta \in \overline{T_{32}} : |u(R_0\mathbf{e}_\theta) - a_2| \le \delta\}.$$

We claim the following hold given we choose suitable γ, δ, ε.

$$0 < \beta_{13}^+ - \beta_{13}^- < \frac{C_\delta}{R_0}, \qquad 0 < \beta_{32}^+ - \beta_{32}^- < \frac{C_\delta}{R_0}; \tag{5.60}$$

$$|u(R_0\mathbf{e}_\theta) - a_j| > \delta \quad \text{for any } j, \quad \theta \in (\beta_{13}^-, \beta_{13}^+) \cup (\beta_{32}^-, \beta_{32}^+); \tag{5.61}$$

$$u(R_0\mathbf{e}_\theta) \in \begin{cases} B_\gamma(a_1), & \theta \in (0, \beta_{13}^-), \\ B_\gamma(a_3), & \theta \in (\beta_{13}^+, \beta_{32}^-), \\ B_\gamma(a_2), & \theta \in (\beta_{32}^+, \pi). \end{cases} \tag{5.62}$$

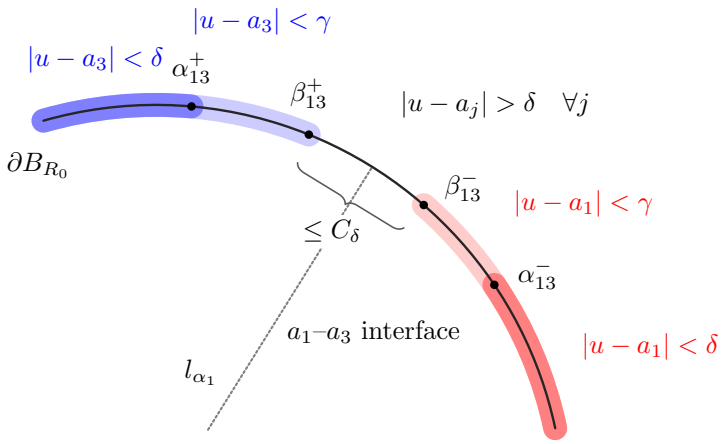

FIGURE 2. Near a_1–a_3 transition on $\partial B_{R_0} \cap \mathbb{R}^2_+$. The transition arc has length $R_0(\beta_{13}^+ - \beta_{13}^-) \le C_\delta$.

Indeed, if $\beta_{13}^+ < \beta_{13}^-$, then by definition for $\theta \in \overline{T_{13}}$, $u(R_0\mathbf{e}_\theta)$ visits a_1–a_3–a_1–a_3 phases in counterclockwise order. Thus $A(R_0) \ge 3\sigma_{13} + \sigma_{32} - o(1)$ which contradicts $A(R_0) < \sigma_{13} + \sigma_{32} + \varepsilon$. For the same reason, for $\theta \in (\beta_{13}^-, \beta_{13}^+)$, $u(R_0\mathbf{e}_\theta)$ cannot visit the δ-neighborhood of any additional phase a_i simply because that will create an additional phase transition and contradict the upper bound. The same argument works for the a_2-a_3 interface. Thus we have verified (5.61). Consider the integral of W over the small arc $\{R_0\mathbf{e}_\theta : \theta \in (\beta_{13}^-, \beta_{13}^+)\}$,

$$C \ge \int_{\{R_0\mathbf{e}_\theta : \theta \in (\beta_{13}^-, \beta_{13}^+)\}} W(u(z))\, d\mathcal{H}^1 = R_0 \int_{\beta_{13}^-}^{\beta_{13}^+} W(u(R_0\mathbf{e}_\theta))\, d\theta \ge CR_0(\beta_{13}^+ - \beta_{13}^-)\delta^2,$$

which gives (5.60) after choosing $C_\delta = \frac{C}{\delta^2}$.

To show (5.62), it suffices to notice that the arcs $\{R_0\mathbf{e}_\theta : \theta \in (\beta_{13}^-, \beta_{13}^+)\}$ and $\{R_0\mathbf{e}_\theta : \theta \in (\beta_{32}^-, \beta_{32}^+)\}$ already carry at least energy $\sigma_{13} + \sigma_{32} - C\delta^2$. If outside these arcs on $\partial B_{R_0} \cap \mathbb{R}^2_+$, u leaves the γ-neighborhood of a phase, that will cause at least $c\gamma^2$ additional energy when the distance to a particular energy well increase from δ to γ, given that $\delta \ll \gamma$. This leads to a contradiction with $A(R_0) \le \sigma_{13} + \sigma_{32} + \varepsilon$, if we choose δ and ε so that $\varepsilon + C\delta^2 < c\gamma^2$. Thus we have verified (5.60)–(5.62). See Figure 2 for an illustration of the choice of angles and the behavior of u near one interface.

Further define

$$\beta_{13} := \frac{\beta_{13}^- + \beta_{13}^+}{2}, \qquad \beta_{32} := \frac{\beta_{32}^- + \beta_{32}^+}{2},$$
$$v(R_0\mathbf{e}_\theta) := a_1\chi_{(0,\beta_{13})}(\theta) + a_3\chi_{(\beta_{13},\beta_{32})}(\theta) + a_2\chi_{(\beta_{32},\pi)}(\theta),$$

where $v \in \mathrm{BV}(\partial B_{R_0} \cap \mathbb{R}^2_+; \{a_1, a_2, a_3\})$. Then (5.60)–(5.62) imply that

$$\|u - v\|_{L^2(\partial B_{R_0} \cap \mathbb{R}^2_+)} \le C(W, M), \tag{5.63}$$

whenever R (and therefore R_0) is sufficiently large. Also $\beta_{13} \to \alpha_1$ and $\beta_{32} \to \alpha_2$ as $R \to \infty$ due to Lemma 5.1.

Recall that U_{ij} denotes the heteroclinic connection connecting a_i to a_j defined in (H_2). Set

$$\nu_{13} = (-\sin\beta_{13}, \cos\beta_{13}), \quad \nu_{32} = (-\sin\beta_{32}, \cos\beta_{32}),$$
$$\mathcal{D}_{13} := \{r\mathbf{e}_\theta : r \in (0, R_0),\, \theta \in (0, \varphi_3)\}, \quad \mathcal{D}_{32} := \{r\mathbf{e}_\theta : r \in (0, R_0),\, \theta \in (\varphi_3, \pi)\}.$$

For sufficiently large R_0, define

$$\tilde{V}(z) := U_{13}(z \cdot \nu_{13}) + U_{32}(z \cdot \nu_{32}) - a_3.$$

We first estimate $J(\tilde{V}, B^+_{R_0})$. Since $\mathcal{D}_{13}$ is angularly separated from $l_{\beta_{32}}$, the exponential decay of U_{32} implies

$$|U_{32}(z \cdot \nu_{32}) - a_3| + |\nabla U_{32}(z \cdot \nu_{32})| + |\nabla^2 U_{32}(z \cdot \nu_{32})| \le Ce^{-c|z|}, \quad z \in \mathcal{D}_{13}.$$

This implies

$$J(\tilde{V}, \mathcal{D}_{13}) \le J(U_{13}(z \cdot \nu_{13}), \mathcal{D}_{13}) + C,$$

since the contribution from $U_{32}(z\cdot\nu_{32}) - a_3$ is exponentially small and can be absorbed by a uniform constant independent of R. On $\mathcal{D}_{13}$, using the coordinates $z = s\mathbf{e}_{\beta_{13}} + t\nu_{13}$, the line segment parallel to $\mathbf{e}_{\beta_{13}}$ with fixed t has length at most $R_0 + |t\cot\beta_{13}|$. This implies

$$J(U_{13}(z \cdot \nu_{13}), \mathcal{D}_{13}) \le R_0\sigma_{13} + |\cot\beta_{13}| \int_{\mathbb{R}} |t| \left(\frac{1}{2}|U_{13}'(t)|^2 + W(U_{13}(t))\right) dt \le R_0\sigma_{13} + C,$$

where we used the fact that $\frac{1}{2}|U_{13}'(t)|^2 + W(U_{13}(t))$ decays exponentially at both ends and therefore is integrable after multiplying $|t\cot(\beta_{13})|$. Consequently,

$$J(\tilde{V}, \mathcal{D}_{13}) \le \sigma_{13}R_0 + C.$$

A similar argument applies to $\mathcal{D}_{32}$, where the major contribution comes from $U_{32}(z\cdot\nu_{32})$, and leads to the estimate

$$J(\tilde{V}, B^+_{R_0}) \le (\sigma_{13} + \sigma_{32})R_0 + C,$$

for some constant depending on u, W and uniform for large R.

Choose a fixed smooth cutoff $0 \le \phi \le 1$ with $\phi(0) = 0$ and $\phi(y) = 1$ for $y \ge 1$, and set

$$\hat{V}(x, y) := \phi(y)\tilde{V}(x, y) + (1 - \phi(y))u_0(x).$$

Note that since $\hat{V}$ is uniformly bounded, (H_1) implies $W(\hat{V}(z)) \le C(W,M)|\hat{V}-a_i|^2$ for any index i. We have

$$\begin{aligned} J(\hat{V}, B_{R_0}^+ \cap \{0<y<1\}) &\le C\bigg[\int_{B_{R_0}^+\cap\{0<y<1\}} |\nabla \tilde{V}|^2 \, dz + \int_{\mathbb{R}} |u_0'|^2 + |u_0 - (a_1\chi_{x>0} + a_2\chi_{x<0})|^2 \, dx \\ &\qquad + \int_{B_{R_0}^+\cap\{0<y<1\}} |\hat{V}(x,y) - (a_1\chi_{x>0} + a_2\chi_{x<0})|^2 \, dxdy \bigg] \\ &\le C(W,u) \end{aligned}$$

for sufficiently large R_0. The last inequality follows from (1.6), (1.7) and exponential decay of $\tilde{V}$ to the corresponding boundary phases.

The construction also implies

$$\begin{aligned} &\hat{V}(x,0) = u_0(x), \qquad J(\hat{V}, B_{R_0}^+) \le (\sigma_{13}+\sigma_{32})R_0 + C, \\ &J(\hat{V}, A_{R_0-1,R_0}^+) + \|\partial_T \hat{V}\|^2_{L^2(\partial B_{R_0}\cap\mathbb{R}^2_+)} + \|\hat{V}-v\|^2_{L^2(\partial B_{R_0}\cap\mathbb{R}^2_+)} \le C. \end{aligned} \tag{5.64}$$

Define the final competitor by

$$V(r\mathbf{e}_\theta) = \begin{cases} \hat{V}(r\mathbf{e}_\theta), & 0<r<R_0-1, \\ \hat{V}(r\mathbf{e}_\theta) + (r-R_0+1)(u(R_0\mathbf{e}_\theta) - \hat{V}(R_0\mathbf{e}_\theta)), & R_0-1 \le r \le R_0. \end{cases}$$

Thus V matches u on both the wall and the semicircle. We compute

$$J(V, A_{R_0-1,R_0}^+) \le C\left(J(\hat{V}, A_{R_0-1,R_0}^+) + \|u-\hat{V}\|^2_{L^2(\partial B_{R_0}\cap\mathbb{R}^2_+)} + \|\partial_T(u-\hat{V})\|^2_{L^2(\partial B_{R_0}\cap\mathbb{R}^2_+)} \right) \le C,$$

which further implies

$$J(u, B_{R_0}^+) \le J(V, B_{R_0}^+) \le (\sigma_{13}+\sigma_{32})R_0 + C.$$

Finally, for sufficiently large R, choose such $R_0 \in (R, 2R)$, and use the lower bound on the annular region:

$$J(u, B_R^+) = J(u, B_{R_0}^+) - J(u, A_{R,R_0}^+) \le (\sigma_{13}+\sigma_{32})R + C(W,u).$$

This completes the proof of the upper bound. □

Next we focus on the a_1-a_3 interface l_{α_1} and establish (1.15) for $\alpha = \alpha_1$ and $(i,j) = (1,3)$. The same argument holds for a_2-a_3 interface l_{α_2}.

Consider the sectors

$$\mathcal{S}_{13} := \{r\mathbf{e}_\theta : r>0,\, \varphi_1 < \theta < \varphi_3\}, \qquad \mathcal{S}_{32} := \{r\mathbf{e}_\theta : r>0,\, \varphi_3 < \theta < \varphi_2\}$$

where the angles are defined in (5.55). Using Lemma 5.2 we obtain that

$$\sigma_{13}R - C \le J(u, \mathcal{S}_{13}\cap B_R) \le J(u, B_R^+) - J(u, \mathcal{S}_{32}\cap B_R) \le \sigma_{13}R + C, \tag{5.65}$$

$$\begin{aligned} &\frac{1}{2}\int_{\mathcal{S}_{13}\cap\{|z|>r_0\}} |\partial_r u|^2 \, dz \\ &\le \lim_{R\to\infty} \left[J(u, \mathcal{S}_{13}\cap A_{r_0,R}) - \int_{r_0}^R \int_{\mathcal{S}_{13}\cap\partial B_r} \left(\frac{1}{2}|\partial_T u|^2 + W(u) \right) d\mathcal{H}^1 \, dr \right] \\ &\le \lim_{R\to\infty} (\sigma_{13}R + C - \int_{r_0}^R (\sigma_{13} - Ce^{-cr}) \, dr) \le C, \end{aligned} \tag{5.66}$$

where the lower bounds of $J(u, \mathcal{S}_{13}\cap B_R)$, $J(u, \mathcal{S}_{32}\cap B_R)$, and the tangential energy cost in $\mathcal{S}_{13}$ follow from the exponential decay estimate on the boundary rays (5.57).

Recall the stress-energy tensor

$$T_{ij} := \partial_i u \cdot \partial_j u - \delta_{ij}\left(\frac{1}{2}|\nabla u|^2 + W(u)\right), \qquad \operatorname{div} T = 0.$$

Set

$$\mathcal{F}_{13}(r) := \int_{\partial B_r \cap \mathcal{S}_{13}} -T\nu \, d\mathcal{H}^1.$$

The stress tensor T on $\partial\mathcal{S}_{13} = l_{\varphi_1} \cup l_{\varphi_3}$ is $O(e^{-cr})$ by Lemma 5.1. Thus, for any $t > s > r_0$,

$$|\mathcal{F}_{13}(t) - \mathcal{F}_{13}(s)| \le C\int_s^t e^{-cr}\, dr \le Ce^{-cs}. \tag{5.67}$$

Thus $\mathcal{F}_{13}(r)$ forms a Cauchy sequence and has a limit as r tends to ∞. We have

$$\frac{1}{R}\int_R^{2R} \mathcal{F}_{13}(r)\, dr = \frac{1}{R}\int_{\mathcal{S}_{13}\cap A_{R,2R}} \left(-(\nabla u)^T \partial_r u + (\frac{1}{2}|\nabla u|^2 + W(u))\nu\right) dz$$

We can apply the estimates in (4.52) and (4.53) to identify the limit

$$\lim_{r\to\infty} \mathcal{F}_{13}(r) = \sigma_{13}\mathbf{e}_{\alpha_1}, \tag{5.68}$$

which together with the convergence rate (5.67) gives

$$\mathcal{F}_{13}(r) = \sigma_{13}\mathbf{e}_{\alpha_1} + O(e^{-cr}). \tag{5.69}$$

Consider an orthogonal change of coordinates $z = x\mathbf{e}_{\alpha_1} + y\nu_{\alpha_1}$ ($\nu_{\alpha_1} = \mathbf{e}_{\alpha_1}^{\perp}$) and define

$$v(x,y) = u(x\mathbf{e}_{\alpha_1} + y\nu_{\alpha_1}) = u(x\cos\alpha_1 - y\sin\alpha_1,\ x\sin\alpha_1 + y\cos\alpha_1).$$

Set

$$k_1 := \tan\frac{\alpha_1}{2}, \qquad k_3 := \tan\frac{\alpha_2 - \alpha_1}{2}, \qquad \mathcal{S} = \{(x,y) : x > 0, -k_1 x < y < k_3 x\}.$$

Then (5.57) becomes the following estimate on $\partial\mathcal{S}$,

$$\begin{aligned} &|v(x,k_3x) - a_3| + |\nabla v(x,k_3x)| + |\nabla^2 v(x,k_3x)| \le Ce^{-cx},\\ &|v(x,-k_1x) - a_1| + |\nabla v(x,-k_1x)| + |\nabla^2 v(x,-k_1x)| \le Ce^{-cx}, \end{aligned} \tag{5.70}$$

for sufficiently large x.

For convenience, we extend our definition of v from $\mathcal{S}$ to the half-plane $\{x > 0\}$. The extended function, still denoted by $v(x,y)$, satisfies

$$v(x,y) = \begin{cases} v(x,y), & (x,y) \in \mathcal{S},\\ a_3, & y > k_3x + 1\\ a_1, & y < -k_1x - 1\\ \text{smooth interpolation between } a_3 \text{ and } v(x,k_3x), & k_3x < y < k_3x + 1,\\ \text{smooth interpolation between } a_1 \text{ and } v(x,-k_1x), & -k_1x - 1 < y < -k_1x. \end{cases}$$

By (5.70),

$$\int_{\mathbb{R}\setminus[-k_1x,k_3x]} (|\partial_x v|^2 + |\partial_y v|^2 + W(v))\, dy \le Ce^{-cx}, \quad \text{for large } x. \tag{5.71}$$

Lemma 5.3. *Set*

$$\Sigma_x := \{(x,y) : y \in \mathbb{R}\}, \quad G(x) := \int_{\Sigma_x} \left(\frac{1}{2}|\partial_y v|^2 - \frac{1}{2}|\partial_x v|^2 + W(v)\right) dy, \quad H(x) := \int_{\Sigma_x} \partial_x v \cdot \partial_y v \, dy.$$

There exist constants C, c, $x_0 > 0$, depending on W and u, such that for every $x > x_0$,

$$|G(x) - \sigma_{13}| + |H(x)| \le Ce^{-cx}. \tag{5.72}$$

$$\int_{x_0}^{\infty} \int_{\Sigma_x} |\partial_x v(x,y)|^2 \, dydx < \infty. \tag{5.73}$$

$$\int_{x_0}^{\infty} \left(\int_{\Sigma_x} \left(\frac{1}{2}|\partial_y v(x,y)^2| + W(v(x,y))\right) dy - \sigma_{13} \right) dx < \infty. \tag{5.74}$$

Proof. First we prove (5.73). By (5.71) it suffices to show

$$\int_{\mathcal{S}\cap\{x>x_0\}} |\partial_x v|^2 \, dz < \infty$$

for some $x_0 > 0$, which is equivalent to

$$\int_{\mathcal{S}_{13}\cap\{r>r_0\}} |\partial_{\mathbf{e}_{\alpha_1}} u|^2 \, dz < \infty, \tag{5.75}$$

for some $r_0 > 0$.

Utilizing $\mathbf{e}_{\alpha_1} \cdot \mathbf{e}_\theta > 0$ for any $\theta \in (\varphi_1, \varphi_3)$, and the relation

$$\frac{1}{2}(1 - \mathbf{e}_{\alpha_1} \cdot \mathbf{e}_\theta)|\nabla u|^2 + \partial_{\mathbf{e}_{\alpha_1}} u \cdot \partial_{\mathbf{e}_\theta} u = |\partial_{\mathbf{e}_{\frac{\alpha_1+\theta}{2}}} u|^2,$$

we have

$$\begin{aligned} \left(\frac{1}{2}|\nabla u|^2 + W(u)\right) + \mathbf{e}_{\alpha_1} \cdot T\mathbf{e}_\theta &= \frac{1}{2}(1 - \mathbf{e}_{\alpha_1} \cdot \mathbf{e}_\theta)|\nabla u|^2 + \partial_{\mathbf{e}_{\alpha_1}} u \cdot \partial_{\mathbf{e}_\theta} u + (1 - \mathbf{e}_{\alpha_1} \cdot \mathbf{e}_\theta)W(u) \\ &= \frac{|\partial_{\mathbf{e}_{\alpha_1}} u + \partial_r u|^2}{|\mathbf{e}_{\alpha_1} + \mathbf{e}_\theta|^2} + (1 - \mathbf{e}_{\alpha_1} \cdot \mathbf{e}_\theta)W(u) \\ &\ge \frac{1}{4}|\partial_{\mathbf{e}_{\alpha_1}} u + \partial_r u|^2 \ge 0. \end{aligned} \tag{5.76}$$

Utilizing (5.65) and (5.69), for fixed r_0 and arbitrarily large R,

$$\begin{aligned} &\int_{\mathcal{S}_{13}\cap A_{r_0,R}} \left(\left(\frac{1}{2}|\nabla u|^2 + W(u)\right) + \mathbf{e}_{\alpha_1} \cdot T\mathbf{e}_\theta \right) dz \\ &= J(u, \mathcal{S}_{13} \cap A_{r_0,R}) - \int_{r_0}^{R} \mathbf{e}_{\alpha_1} \cdot \mathcal{F}_{13}(r) \, dr \\ &\le J(u, \mathcal{S}_{13} \cap B_R^+) - \sigma_{13}(R - r_0) + C \le C. \end{aligned} \tag{5.77}$$

Moreover, (5.76) implies

$$|\partial_{\mathbf{e}_{\alpha_1}} u|^2 \le 2\left(|\partial_{\mathbf{e}_{\alpha_1}} u + \partial_r u|^2 + |\partial_r u|^2\right) \le 8\left(\left(\frac{1}{2}|\nabla u|^2 + W(u)\right) + \mathbf{e}_{\alpha_1} \cdot T\mathbf{e}_\theta\right) + 2|\partial_r u|^2.$$

Combining this with (5.77) and (5.66) leads to (5.75), and further (5.73).

We continue in the original coordinates. Take $\kappa := \sqrt{1+k_1^2} + \sqrt{1+k_3^2}$ and apply the divergence theorem to T over the domain

$$\mathcal{S}_{13} \cap B_{\kappa r} \cap \{z : z \cdot \mathbf{e}_{\alpha_1} > r\}, \quad r > r_0,$$

whose boundary consists of four parts:

$$B_1 := \{r\mathbf{e}_{\alpha_1} + s\nu_{\alpha_1} : -k_1 r < s < k_3 r\}, \quad B_2 := \{s\mathbf{e}_{\varphi_1} : \sqrt{1+k_1^2}r < s < \kappa r\},$$
$$B_3 := \{s\mathbf{e}_{\varphi_3} : \sqrt{1+k_3^2}r < s < \kappa r\}, \quad B_4 := \{\kappa r\mathbf{e}_\theta : \varphi_1 < \theta < \varphi_3\},$$

to get

$$\int_{B_1} -T\mathbf{e}_{\alpha_1}\, d\mathcal{H}^1 + \int_{B_2} -T\nu_{\varphi_1}\, d\mathcal{H}^1 + \int_{B_3} T\nu_{\varphi_3}\, d\mathcal{H}^1 + \int_{B_4} T\nu\, d\mathcal{H}^1 = 0. \tag{5.78}$$

By (5.57), the two radial sides B_2 and B_3 only contribute $O(e^{-cr})$, so (5.69) implies

$$\int_{B_1} -T\mathbf{e}_{\alpha_1}\, d\mathcal{H}^1 = \sigma_{13}\mathbf{e}_{\alpha_1} + O(e^{-cr}).$$

Its tangential and normal components are

$$\begin{aligned} \mathbf{e}_{\alpha_1} \cdot \int_{B_1} -T\mathbf{e}_{\alpha_1}\, d\mathcal{H}^1 &= \int_{-k_1 r}^{k_3 r} \left(\frac{1}{2}|\partial_y v|^2 - \frac{1}{2}|\partial_x v|^2 + W(v)\right) dy = \sigma_{13} + O(e^{-cr}), \\ \nu_{\alpha_1} \cdot \int_{B_1} -T\mathbf{e}_{\alpha_1}\, d\mathcal{H}^1 &= \int_{-k_1 r}^{k_3 r} -(\partial_y v \cdot \partial_x v)\, dy = O(e^{-cr}). \end{aligned} \tag{5.79}$$

This together with (5.70) implies (5.72). Finally, (5.74) follows from (5.72) and (5.73) immediately. □

Let

$$\mathcal{U} := \{U_{13}(\cdot - h) : h \in \mathbb{R}\}, \quad \text{be the set of all translates of } U_{13}.$$

$$\mathcal{A} := \{w \in H^1_{loc}(\mathbb{R}, \mathbb{R}^2) : w(-\infty) = a_1,\, w(+\infty) = a_3,\, \int_{\mathbb{R}} (\frac{1}{2}|w'|^2 + W(w))\, dx < \infty\}.$$

For simplicity, we shall henceforth write U_{13} as U. We further define

$$d_2(w, \mathcal{U}) := \inf_{\tau \in \mathcal{U}} \|w - \tau\|_{L^2(\mathbb{R};\mathbb{R}^2)}.$$

We invoke the following result, most of which comes from Schatzman's work [38], to show that for most $x > 0$, $v|_{\Sigma_x}$, which belongs to $\mathcal{A}$, can be approximated by a translation of U.

Lemma 5.4. *Consider $w \in \mathcal{A}$ such that $\|w\|_{L^\infty} + \|w'\|_{L^\infty} \le M$. For any $\varepsilon > 0$, there exists $\eta_\varepsilon > 0$ such that*

$$\int_{\mathbb{R}} \left(\frac{1}{2}|w'|^2 + W(w)\right) dx - \sigma_{13} \le \eta_\varepsilon \quad \Rightarrow \quad d_2(w, \mathcal{U}) \le \varepsilon. \tag{5.80}$$

Moreover, there exists $\varepsilon_0 = \varepsilon_0(W, M)$ such that if $d_2(w, \mathcal{U}) < \varepsilon_0$, then there exists a unique $h = h(w)$ such that

$$d_2(w, \mathcal{U}) = \|w - U(\cdot - h)\|_{L^2},$$

and

$$\int_{\mathbb{R}} \left(\frac{1}{2}|w'|^2 + W(w)\right) dx - \sigma_{13} \ge c\|w - U(\cdot - h)\|^2_{H^1(\mathbb{R})}, \tag{5.81}$$

for some $c = c(W) > 0$. Moreover, $h(w)$ is a function of class C^1 in the affine L^2 space.

Proof. The existence, uniqueness, and differentiability of $h(w)$ follow directly from [38, Lemma 2.1]. (5.80) follows from an application of [38, Theorem 3.1 and Corollary 3.2]. Note that the original results in [38] are stated in a two-well setting. Under (H_3) or (H_3'), the strict triangle inequality exclude splitting by another phase. Hence we can adapt the compactness argument to the a_1-a_3 connection. Hypothesis (H_4) then identifies the set of minimizers with $\mathcal{U}$.

For the coercivity (5.81), we follow closely the proof of [38, Lemma 4.5]. Indeed, denoting by $q = w - U(\cdot - h)$, one can expand the 1D energy

$$\int_{\mathbb{R}} \left(\frac{1}{2}|w'|^2 + W(w) \right) dx - \sigma_{13} = \frac{1}{2} \int_{\mathbb{R}} \left(|q'|^2 + q \cdot D^2 W(U(y-h))q \right) dy + \mathcal{R} \geq c\|q\|_{H^1}^2 + \mathcal{R},$$

where we used (H_4) in the last inequality. The remainder term $\mathcal{R}$ has the explicit form

$$\mathcal{R} := \int_{\mathbb{R}} \int_0^1 (1-s)\, q(y) \cdot \left[D^2 W(U(y-h) + sq(y)) - D^2 W(U(y-h)) \right] q(y)\, ds\, dy.$$

The elementary estimate

$$|q(x)|^3 \leq \int_{-\infty}^{x} 3|q(t)|^2 |q(t)|'\, dt \leq 3\|q'\|_{L^\infty} \|q\|_{L^2}^2 \leq C(M,W)\|q\|_{L^2}^2 \tag{5.82}$$

implies that

$$|\mathcal{R}| \leq C(M,W)\|q\|_{L^2}^{\frac{8}{3}}.$$

Thus the remainder term can be absorbed by the coercive term $c\|q\|_{H^1}^2$. □

Define the "good" set

$$\mathcal{G} := \{x > 0 : d_2(v(x,\cdot), \mathcal{U}) < \varepsilon_0\}.$$

From Lemma 5.4, there exists a function $h \in C^1(\mathcal{G}; \mathbb{R})$ such that

$$d_2(v(x,\cdot), \mathcal{U}) = \|v(x,\cdot) - U(\cdot - h(x))\|_{L^2}, \quad x \in \mathcal{G}.$$

We have the following identities:

$$\int_{\Sigma_x} (v - U(y - h(x))) \cdot U'(y - h(x))\, dy = 0, \quad x \in \mathcal{G}, \tag{5.83}$$

$$h'(x) = -\frac{\int_{\Sigma_x} \partial_x v \cdot U'(y-h(x))\, dy}{\int_{-\infty}^{\infty} \left[|U'(y-h(x))|^2 - U''(y-h(x)) \cdot (v - U(y-h(x))) \right] dy}. \tag{5.84}$$

Identity (5.83) follows from the fact that $U(\cdot - h(x))$ minimizes the L^2 distance to $v(x,\cdot)$. And (5.84) follows from differentiating (5.83) with respect to x.

Using (5.72), (5.73), (5.74), (5.80) and (5.81), we have for $x \in \mathcal{G}$,

$$\begin{aligned}
&\left| \int_{\Sigma_x} \partial_x v \cdot U'(y-h(x))\, dy \right| \\
\leq &\left| \int_{\Sigma_x} \partial_x v \cdot (U'(y-h(x)) - \partial_y v)\, dy \right| + \left| \int_{\Sigma_x} \partial_x v \cdot \partial_y v\, dy \right| \\
\leq &\|\partial_x v(x,\cdot)\|_{L^2} \cdot \|v(x,\cdot) - U(\cdot - h(x))\|_{H^1} + Ce^{-cx} \\
\leq &\frac{1}{2}\|\partial_x v(x,\cdot)\|_{L^2}^2 + C\Big(\int_{\Sigma_x} \frac{1}{2}|\partial_y v|^2 + W(v)\, dy - \sigma_{13} \Big) + Ce^{-cx} \in L^1(\mathcal{G}).
\end{aligned} \tag{5.85}$$

Proposition 5.5. *There exists $h_{13} \in \mathbb{R}$ such that*

$$\lim_{x\to+\infty} \|v(x,\cdot) - U_{13}(\cdot - h_{13})\|_{L^2(\mathbb{R})} = 0. \tag{5.86}$$

For every $T > 0$ and $\beta \in (0,1)$, the above convergence can be upgraded to

$$\lim_{x\to\infty} \|v(x,\cdot) - U_{13}(\cdot - h_{13})\|_{C^{2,\beta}([-T,T])} = 0. \tag{5.87}$$

The same conclusion holds for l_{α_2} with U_{32} and a finite shift $h_{32} \in \mathbb{R}$.

Proof. Let $d_2(x) := d_2(v(x,\cdot),\mathcal{U})$. For $x_2 > x_1 > x_0$,

$$\begin{aligned} |d_2(x_2) - d_2(x_1)| &\le \|v(x_2,\cdot) - v(x_1,\cdot)\|_{L^2} \\ &\le \int_{x_1}^{x_2} \Big(\int_{\Sigma_x} |\partial_x v|^2 \, dy\Big)^{\frac{1}{2}} dx \le \sqrt{x_2 - x_1} \left(\int_{x_0}^{\infty}\int_{\Sigma_x} |\partial_x v|^2 \, dy\, dx\right)^{1/2}. \end{aligned}$$

Thus d_2 is uniformly continuous. If $d_2(x) \not\to 0$, there exist $\varepsilon, \delta > 0$ and disjoint intervals $\{I_n\}_{n=1}^{\infty}$ such that

$$|I_n| \ge \delta, \qquad d_2(x) > \varepsilon \ \text{ on } I_n. \tag{5.88}$$

Using (5.80) we infer that

$$\int_{x_0}^{\infty} \left[\int_{\Sigma_x} \left(\frac{1}{2}|\partial_y v|^2 + W(v)\right) dy - \sigma_{13}\right] dx \ge \sum_n \delta\eta_\varepsilon = \infty,$$

which yields a contradiction with (5.74). Hence $d_2(x) \to 0$.

After enlarging x_0 if necessary, we can assume $(x_0,\infty) \subset \mathcal{G}$. The denominator in (5.84) is bounded below by

$$\|U'\|_{L^2}^2 - d_2(x)\|U''\|_{L^2} \ge \frac{1}{2}\|U'\|_{L^2}^2, \quad \forall\, x > x_0.$$

This, combined with the estimate (5.85) for the numerator, gives $|h'(x)| \in L^1(x_0,\infty)$. It follows that there exists $h_{13} \in \mathbb{R}$ such that $h(x) \to h_{13}$ and

$$\|v(x,\cdot) - U(\cdot - h_{13})\|_{L^2} \le d_2(x) + \|U(\cdot - h(x)) - U(\cdot - h_{13})\|_{L^2} \ \to\ 0,$$

which proves (5.86). The uniform bound on $|\nabla v|$ and estimate (5.82) imply

$$\|v(x,\cdot) - U(\cdot - h_{13})\|_{L^\infty} \to 0. \tag{5.89}$$

Finally, interior elliptic estimates can upgrade the uniform convergence to $C^{2,\beta}_{loc}$ convergence for any $\beta \in (0,1)$, which gives (5.87).

□

Proof of Theorem 1.4. Proposition 5.5 proves the asserted heteroclinic convergence (1.15) along interior sharp interfaces. It remains to establish the localization of the diffuse interface (1.14).

Fix a sufficiently small $\gamma > 0$ and set the diffuse interface as in (1.13). For both interfaces l_{α_1} and l_{α_2}, (5.89) implies there exists C_γ such that

$$D_\gamma \cap (\mathcal{S}_{13} \cup \mathcal{S}_{32}) \subset \{z : \operatorname{dist}(z, (l_{\alpha_1} \cup l_{\alpha_2})) \le C_\gamma\}. \tag{5.90}$$

To prove (1.14), it remains to show that there exists a sufficiently large r_0 such that D_γ does not intersect with $\{r\mathbf{e}_\theta : r > r_0,\ \theta \in (0,\varphi_1) \cup (\varphi_2,\pi)\}$. We focus on

$$\mathcal{S}_1 := \{r\mathbf{e}_\theta : r > 0,\ \theta \in (0,\varphi_1)\}.$$

By a similar argument as in the proof of Lemma 5.2, there exists r_0 such that for any $r > r_0$, we can find a good radius $R \in (r, 2r)$ such that

$$|u(R\mathbf{e}_\theta) - a_1| \le \frac{\gamma}{2}, \quad \theta \in (0, \varphi_1).$$

The idea is that on the semicircle $\partial B_R \cap \mathbb{R}^2_+$, if the total energy (only counting tangential derivative and potential) is sufficiently close to $\sigma_{13} + \sigma_{32}$, then most of the energy should be concentrated near $l_{\alpha_1} \cup l_{\alpha_2}$, preventing u from escaping a small neighborhood of energy wells away from the sharp interface as it will cost extra energy.

We may further assume, by enlarging r_0 if necessary, that for every $r > r_0$,

$$|u(r\mathbf{e}_{\varphi_1}) - a_1| \le \frac{\gamma}{2}, \quad |u_0(r) - a_1| \le \frac{\gamma}{2}.$$

Thus for any two of these good radii $R_1 < R_2$, the annular region $\mathcal{S}_{R_1,R_2} := \{r\mathbf{e}_\theta : r \in (R_1, R_2),\ \theta \in (0, \varphi_1)\}$ satisfies $|u - a_1| \le \frac{\gamma}{2}$ on $\partial\mathcal{S}_{R_1,R_2}$, and Lemma 2.2 therefore implies

$$|u - a_1| \le \frac{\gamma}{2}, \quad \text{in } \mathcal{S}_{R_1,R_2}.$$

Taking $R_2 \to \infty$, we obtain

$$(\mathcal{S}_1 \setminus B_r) \cap D_\gamma = \emptyset, \quad \text{for some } r > 0.$$

Similarly, we have

$$(\{r\mathbf{e}_\theta : \theta \in (\varphi_2, \pi)\} \setminus B_r) \cap D_\gamma = \emptyset, \quad \text{for some } r > 0.$$

Hence we have proved Theorem 1.4 for the case of two interior interfaces. The same argument applies to the single interior interface case.

□

References

[1] Alikakos, N. D., and Fusco, G. Density estimates for vector minimizers and applications. *Discrete & Continuous Dynamical Systems 35*, 12 (2015), 5631–5663.

[2] Alikakos, N. D., and Fusco, G. A maximum principle for systems with variational structure and an application to standing waves. *Journal of the European Mathematical Society 17*, 7 (2015), 1547–1567.

[3] Alikakos, N. D., and Fusco, G. Sharp lower bounds for the vector Allen–Cahn energy and qualitative properties of minimizers under no symmetry hypotheses. *Bulletin of the Hellenic Mathematical Society 67* (2023), 12–58.

[4] Alikakos, N. D., Fusco, G., and Smyrnelis, P. *Elliptic systems of phase transition type*. Springer, 2018.

[5] Alikakos, N. D., and Geng, Z. On the triple junction problem without symmetry hypotheses. *Archive for Rational Mechanics and Analysis 248*, 2 (2024), 24.

[6] Ambrosio, L., and Cabré, X. Entire solutions of semilinear elliptic equations in $\mathbb{R}^3$ and a conjecture of De Giorgi. *Journal of the American Mathematical Society 13*, 4 (2000), 725–739.

[7] Baldo, S. Minimal interface criterion for phase transitions in mixtures of Cahn-Hilliard fluids. *Annales de l'Institut Henri Poincaré C, Analyse non linéaire 7*, 2 (1990), 67–90.

[8] Bethuel, F. Asymptotics for 2-dimensional vectorial Allen-Cahn systems. *Acta Mathematica 234*, 2 (2025), 189–314.

[9] Bronsard, L., Gui, C., and Schatzman, M. A three-layered minimizer in $\mathbb{R}^2$ for a variational problem with a symmetric three-well potential. *Communications on Pure and Applied Mathematics 49*, 7 (1996), 677–715.

[10] Chan, H., and Wei, J. On De Giorgi's conjecture: recent progress and open problems. *Science China Mathematics 61* (2018), 1925–1946.

[11] De Giorgi, E. Convergence problems for functionals and operators. In *Proc. Int. Meeting on Recent Methods in Nonlinear Analysis* (1979), Pitagora, Bologna, pp. 131–188.

[12] Del Pino, M., Kowalczyk, M., and Wei, J. On De Giorgi's conjecture in dimension $n \ge 9$. *Annals of Mathematics 174*, 3 (2011), 1485–1569.

[13] Du, W., Mooney, C., Yang, Y., and Zhu, J. A half-space Bernstein theorem for anisotropic minimal graphs. *Journal of the European Mathematical Society* (2025). Published online first.

[14] Du, W., Wang, L., and Yang, Y. Flat level sets of Allen-Cahn equation in half-space. *arXiv preprint arXiv:2412.20335* (2024).
[15] Edelen, N., and Wang, Z. A Bernstein-type theorem for minimal graphs over convex domains. *Annales de l'Institut Henri Poincaré C 39*, 3 (2022), 749–760.
[16] Farina, A., and Valdinoci, E. Flattening results for elliptic PDEs in unbounded domains with applications to overdetermined problems. *Archive for Rational Mechanics and Analysis 195*, 3 (2010), 1025–1058.
[17] Farina, A., and Valdinoci, E. 1D symmetry for solutions of semilinear and quasilinear elliptic equations. *Transactions of the American Mathematical Society 363*, 2 (2011), 579–609.
[18] Flores, G., Padilla, P., and Tonegawa, Y. Higher energy solutions in the theory of phase transitions: a variational approach. *Journal of Differential Equations 169*, 1 (2001), 190–207.
[19] Florit-Simon, E., and Serra, J. On stable solutions to the Allen-Cahn equation with bounded energy density in $\mathbb{R}^4$. *arXiv preprint arXiv:2509.02739* (2025).
[20] Fonseca, I., and Tartar, L. The gradient theory of phase transitions for systems with two potential wells. *Proceedings of the Royal Society of Edinburgh Section A: Mathematics 111*, 1-2 (1989), 89–102.
[21] Fusco, G. Layered solutions to the vector Allen–Cahn equation in $\mathbb{R}^2$. minimizers and heteroclinic connections. *Communications on Pure and Applied Analysis 16*, 5 (2017), 1807–1841.
[22] Fusco, G. Minimizing under relaxed symmetry constraints: Triple and N-junctions. *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze 25*, 2 (2024), 887–940.
[23] Fusco, G. Connectivity of the diffuse interface and fine structure of minimizers in the Allen–Cahn theory of phase transitions. *São Paulo Journal of Mathematical Sciences 19*, 1 (2025), 14.
[24] Gazoulis, D. On the Γ-convergence of the Allen–Cahn functional with boundary conditions. *Proceedings of the Royal Society of Edinburgh Section A: Mathematics 155*, 5 (2025), 1557–1579.
[25] Geng, Z. Rigidity results for a triple junction solution of the Allen-Cahn system. *SIAM Journal on Mathematical Analysis 57*, 5 (2025), 5570–5597.
[26] Geng, Z. Uniqueness of the blow-down limit for a triple junction problem. *Communications on Pure and Applied Mathematics 78*, 2 (2025), 500–534.
[27] Ghoussoub, N., and Gui, C. On a conjecture of De Giorgi and some related problems. *Mathematische Annalen 311*, 3 (1998), 481–491.
[28] Giusti, E. *Minimal Surfaces and Functions of Bounded Variation*, vol. 80 of *Monographs in Mathematics*. Birkhäuser Boston, 1984.
[29] Guaraco, M. A. Min–max for phase transitions and the existence of embedded minimal hypersurfaces. *Journal of Differential Geometry 108*, 1 (2018), 91–133.
[30] Gui, C., and Schatzman, M. Symmetric quadruple phase transitions. *Indiana University Mathematics Journal 57*, 2 (2008), 781–836.
[31] Hamel, F., Liu, Y., Sicbaldi, P., Wang, K., and Wei, J. Half-space theorems for the Allen-Cahn equation and related problems. *Journal für die reine und angewandte Mathematik (Crelles Journal) 2021*, 770 (2021), 113–133.
[32] Liu, Y., Wang, K., and Wei, J. Global minimizers of the Allen-Cahn equation in dimension $n \geq 8$. *Journal de Mathématiques Pures et Appliquées 108*, 6 (2017), 818–840.
[33] Monteil, A., and Santambrogio, F. Metric methods for heteroclinic connections in infinite dimensional spaces. *Indiana University Mathematics Journal 69*, 4 (2020), 1445–1503.
[34] Pacard, F., and Wei, J. Stable solutions of the Allen–Cahn equation in dimension 8 and minimal cones. *Journal of Functional Analysis 264*, 5 (2013), 1131–1167.
[35] Sandier, E., and Sternberg, P. Allen–Cahn solutions with triple junction structure at infinity. *Communications on Pure and Applied Mathematics 77*, 11 (2024), 4163–4211.
[36] Savin, O. Regularity of flat level sets in phase transitions. *Annals of Mathematics 169*, 1 (2009), 41–78.
[37] Savin, O. Minimal surfaces and minimizers of the Ginzburg Landau energy. *Contemporary Mathematics, AMS 528* (2010), 43–57.
[38] Schatzman, M. Asymmetric heteroclinic double layers. *ESAIM: Control, Optimisation and Calculus of Variations 8* (2002), 965–1005.
[39] Sternberg, P. The effect of a singular perturbation on nonconvex variational problems. *Archive for Rational Mechanics and Analysis 101*, 3 (1988), 209–260.
[40] Sternberg, P., and Ziemer, W. P. Local minimisers of a three-phase partition problem with triple junctions. *Proceedings of the Royal Society of Edinburgh Section A: Mathematics 124*, 6 (1994), 1059–1073.
[41] Tonegawa, Y., and Wickramasekera, N. Stable phase interfaces in the van der Waals-Cahn-Hilliard theory. *Journal für die Reine und Angewandte Mathematik 668* (2012), 191–210.

[42] Wang, K. A new proof of Savin's theorem on Allen–Cahn equations. *Journal of the European Mathematical Society 19*, 10 (2017), 2997–3051.
[43] Wang, K., and Wei, J. Finite Morse index implies finite ends. *Communications on Pure and Applied Mathematics 72*, 5 (2019), 1044–1119.
[44] Wang, K., and Wei, J. Second order estimate on transition layers. *Advances in Mathematics 358* (2019), 106856.
[45] White, B. Lecture notes on flat chains and geometric measure theory, Stanford, (notes by Otis Chodosh).